

Hedge Fund Performance Evaluation: A Stochastic Discount Factor Approach

Warren Bailey, Haitao Li, and Xiaoyan Zhang *

Current draft: 6th December 2004

JEL Classifications: G23, G11, G12

Keywords: hedge funds, asset pricing, Hansen-Jagannathan distance

* Bailey, Li, and Zhang are from Johnson Graduate School of Management, Cornell University, Ithaca, NY 14853-6201, and can be reached at wbb1@cornell.edu, hl70@cornell.edu, and xz89@cornell.edu, respectively. We thank Steve Brown, Jin-Chuan Duan, Raymond Kan, Andrew Karolyi, Ernst Schaumburg, and workshop participants at Cornell/Rochester and Toronto for helpful discussions. We thank Rui Zhao and Eric Litton for characteristics data collection. We appreciate financial support from the Q-Group. We are responsible for all remaining errors.

© 2004 Warren Bailey, Haitao Li, and Xiaoyan Zhang.

Hedge Fund Performance Evaluation: A Stochastic Discount Factor Approach

Abstract

We analyze hedge fund performance using the stochastic discount factor (SDF) approach and imposing the arbitrage-free requirement to correctly value the derivatives and dynamic trading strategies used by hedge funds. Using SDFs of many asset-pricing models, we evaluate hedge fund portfolios based on style and characteristics. Without the arbitrage-free requirement, pricing errors are relatively small and a few models can explain hedge fund returns. With this requirement, pricing errors are much bigger, and all models fail to price style and volatility portfolios. Fund manager characteristics like age, experience, and education explain some of the mispricing of our best risk model.

The hedge fund industry has become one of the fastest growing segments of the investment community since the 1990s. According to a hedge fund research company, Tremont, over the past 10 years, the number of hedge funds has increased at an average rate of 18% per year, showing a total growth rate of 400%. Today there are thousands of hedge funds operating, including about 1,000 “funds of funds”. The total amount of assets under management is between \$500 and \$700 billion. In recent years, hedge funds have attracted the attention of many investors, including high-net-worth individuals, banks, endowments and foundations, insurance companies, and pension plan sponsors. With the development of “funds of funds”, hedge funds are increasingly accessible even to retail investors.¹

There are several unique features that distinguish hedge funds from traditional mutual funds and help explain the increasing popularity of hedge funds. First, due to their legal structure, hedge funds are not subject to the same level of regulation as mutual funds and they enjoy greater flexibility in choosing their investment strategies. For example, hedge funds frequently use short selling, leverage, and derivatives, strategies rarely used by ordinary mutual funds, to enhance returns and reduce risk. As a result, hedge funds can better exploit potential market inefficiencies and seek positive abnormal returns. Hedge fund returns are also less correlated with conventional investment vehicles and they provide additional diversification benefits.

The incentive structure of hedge funds can also attract talented portfolio managers to the industry and help resolve the agency problems typically facing mutual fund investors. Today, most hedge funds charge incentive fees, typically 15% to 20% of profits, in addition to a fixed management fee, which is typically 1% to 2% of assets under management. With such an incentive structure, hedge fund managers can expect greater financial rewards than their counterparts in mutual funds. As a result, the hedge fund industry has attracted top talent from traditional financial institutions at an accelerating rate in the last decade. The incentive structure may create stronger motivation for hedge fund managers to generate positive returns because most of their income comes from incentive fees. Moreover, hedge fund managers typically invest a significant part of their personal capital in the funds they manage and may even require key employees to invest as well, ensuring that the interests of all parties are aligned.

¹ For a description of the current state of the hedge fund industry, see the 2003 edition of the Tremont publication “The Case for Hedge Funds” available at http://www.hedgeworld.com/tremont_tass_research/.

Unfortunately the structure of hedge funds is far from perfect and there is a widely recognized dark side to the industry. The flexible investment strategies, especially those involving derivatives and leverage, enable hedge funds to take on excessive risks that could endanger investor capital and even jeopardize the whole financial system.² While rich compensation can attract genuine talent, not every manager entering the industry has real skills.³ Given the limited number of arbitrage opportunities in financial markets, the increased competition among the increasing number of hedge funds may also reduce the profitability to the industry as a whole. Furthermore, even if hedge funds can make positive abnormal returns, it is not clear whether there is any money left on the table for their investors after taking out the rich managerial compensation.

All these considerations raise an interesting question for both hedge fund investors and financial economists. Do hedge funds provide their investors with enough positive abnormal returns to justify the high fees they charge? Alternatively, does the history of hedge fund returns indicate that they have delivered positive net-of-fee abnormal returns to their investors?

Given that investors have many other choices of investment services such as traditional mutual funds, venture capital funds, banks, insurance companies, and exchange traded funds, evaluating the performance of the increasingly popular hedge funds is important for investment decision making. The importance of the issue is further elevated because hedge funds are much less regulated than traditional investment services and there is much less information on hedge fund holdings, performance, and other characteristics. Information on hedge fund performance is also of interest to regulators concerned about threats the industry may pose to the financial system as a whole.

Hedge fund performance evaluation raises interesting challenges for existing asset pricing models. While these models have been reasonably successful in explaining returns on conventional securities such as stocks, bonds, and mutual funds, it is unclear whether they can capture the returns of the highly complicated dynamic trading strategies of hedge funds. If we find specific characteristics of hedge funds (style, specialization, management) are associated with consistently positive abnormal returns, it helps us identify and understand potential

² The widely publicized case of Long Term Capital Management illustrates how a highly leveraged hedge fund can threaten global financial markets.

³ See, for example, Akerlof (1970) on adverse selection issues.

inefficiencies in certain parts of the financial markets. It also serves to reveal the inadequacies of existing asset pricing models.

In this paper, we provide a comprehensive empirical analysis of hedge fund performance evaluation using the stochastic discount factor (SDF) approach. Our paper makes both methodological and empirical contributions to the fast growing literature on hedge funds.

Methodologically, we differ from the existing literature in two aspects. First, we evaluate hedge fund performance using the Hansen and Jagannathan (1997) distance (HJ-distance hereafter) measure within the SDF framework, rather than employing time-series regression approaches.⁴ As elegantly demonstrated by Cochrane (2001), the SDF approach provides a unified framework for asset pricing analysis and has been widely used to study equities, bonds, options, trading strategies, and mutual funds.⁵ Since the HJ-distance measures are defined within the SDF framework, we can easily examine and compare a broad spectrum of models, and present the big picture on hedge fund performance with respect to different risk specifications. Furthermore, the SDF setup and HJ-distance can easily incorporate conditional information, thereby allowing risk premiums to be time varying.

Second, the HJ-distance measures allow us to take into account the arbitrage-free requirement for the SDF (that is, the SDF has to be positive to avoid arbitrage opportunities). This arbitrage-free requirement is particularly relevant for hedge funds. Given their heavy use of derivatives and dynamic trading strategies,⁶ existing performance measures such as the Sharpe ratio and Jensen's alpha are inappropriate. For example, Goetzmann, Ingersoll, Spiegel, and Welch (2003) show that the Sharpe ratio can easily be manipulated by investing in derivatives. Grinblatt and Titman (1989) show that a manager selling a call option on the index will be incorrectly classified as displaying superior performance by an investor using Jensen's alpha.⁷ Wang and Zhang (2004) show that the arbitrage-free requirement for a SDF is important for evaluating asset-pricing models, especially with nonlinear returns and conditional variables. The HJ-distance naturally incorporates the arbitrage-free requirement, while it is extremely difficult

⁴ We are probably the first study that evaluates hedge fund performance using the Hansen-Jagannathan (1997) distance and explicitly imposing the arbitrage-free requirement.

⁵ See, for example, Jagannathan and Wang (1996), Hodrick and Zhang (2001), Buraschi and Jackwerth (2001), Coval and Shumway (2001), Ahn, Conrad, and Dittmar (2003), Knez and Chen (1996), and Ferson, Henry, and Kisgen (2003).

⁶ See, for example, Fung and Hsieh 1997, Agarwal and Naik 2004, and Ben Dor and Jagannathan 2002.

⁷ See also Dybvig and Ingersoll (1982) and Glosten and Jagannathan (1994). Merton (1981) and Dybvig and Ross (1985) also show that funds using dynamic trading strategies exhibit option-like features.

to impose the arbitrage-free requirement in a (time series) linear regression framework as commonly used with standard linear asset pricing models.

With this new empirical approach, we apply a wide range of asset pricing models to evaluate the performance of hedge fund portfolios. Unlike mutual funds that typically invest in traditional equities and bonds, hedge funds invest in a much wider array of instruments and therefore are exposed to additional sources of risk beyond those considered in typical asset pricing models. To empirically capture the unique features of hedge fund returns, we consider the traditional CAPM, a Fama-French three factor model (with market, size, and book-to-market factors), and models with option returns to capture option-like features in hedge fund returns. We also consider empirical pricing kernels retrieved from returns of industry portfolios and several alternative asset classes beyond stocks and bonds.

We use the above models to evaluate the performance of value-weighted hedge fund portfolios. Note that the evaluation exercise is a joint test of model risk specification and portfolio performance. Hedge fund portfolios are constructed by using either the eleven investment styles defined by TASS or by sorting on commonly used characteristics in the asset pricing literature, such as return volatility, size, return momentum, and clustering analysis. The variety of portfolios dissects hedge fund returns in different ways, making the results on performance evaluation more robust to potential manipulations by hedge fund managers in their self-reported styles.

There are four main empirical findings. First, we find significant return spreads among hedge fund portfolios sorted on various characteristics. Second, without the arbitrage-free requirement, most linear asset pricing models are able to explain the returns on characteristic sorted portfolios, but not the style portfolios. Third, with the arbitrage-free requirement, it is much more difficult to price the hedge fund portfolios, which indicates the importance of the arbitrage-free requirement. Some models are still able to price the size, momentum and cluster portfolios, but none of the models can price either style portfolios or volatility portfolios. This indicates that either the models derived from stock and option market factors cannot give adequate risk adjustments for hedge funds, or the style and volatility portfolios outperform. Finally, fund manager characteristics, such as age, experience and educational background, help to explain some of the pricing errors of the best asset pricing model, but not the pricing errors from an empirical kernel defined over hedge fund space.

This paper is organized as follows. Section I discusses the methodology and models we use to evaluate hedge fund performance. Section II describes the dataset and provides summary information on style and characteristic sorted portfolios. Section III presents empirical results. Section IV concludes.

I. Hedge Fund Performance Evaluation Using the Stochastic Discount Factor

A. HJ-distance methodology

Modern asset pricing theory posits that, under the law of one price, there should exist a set of SDFs that correctly prices every single asset (including all tradable assets and dynamic trading strategies) in the economy. Suppose there are n assets traded in the economy whose returns are denoted as $R_i, i=1, \dots, n$. Then we have,

$$(1) \quad E[m_{t+1} R_{i,t+1} | \Omega_t] = p, \quad \forall m_{t+1} \in M_{t+1}, \forall i, t,$$

where m_{t+1} is a member of the set of correct SDFs, M_{t+1} . Ω_t is the information set at time t , and p is the price for the returns ($p=1$, if the returns are gross returns; $p=0$ if the returns are excess returns). The returns are risk adjusted by the discount rate, m_{t+1} , so that the expected present value per dollar invested is equal to today's price. Thus, m_{t+1} is usually called a SDF (or pricing kernel). To avoid possible arbitrage opportunities, the correct SDFs should all be positive. If we explicitly impose this arbitrage-free requirement, we have a new set of correct SDFs, M^+ (a subset of M), and

$$(2) \quad E[m_{t+1}^+ R_{i,t+1} | \Omega_t] = p, \quad m_{t+1}^+ > 0, \forall m_{t+1}^+ \in M_{t+1}^+, \forall i, t.$$

Every asset-pricing model implies a relevant SDF, denoted as y . If a model is correctly specified, it should price all assets in the economy:

$$(3) \quad E[y_{t+1} R_{i,t+1} | \Omega_t] = p, \quad \forall i, t.$$

Then y belongs to the correct SDF set M or M^+ . The first HJ-distance, d , is defined as the minimum distance between the model implied SDF, y , and the set of correct SDFs, M ,

$$(4) \quad d = \min_{m \in M} \sqrt{E(y_t - m_t)^2},$$

If the model is correctly specified, the HJ-distance should be zero. When we explicitly impose the arbitrage-free requirement, we define the second HJ-distance, d^+ , to be the minimum distance between y and the set of correct and non-negative SDFs, M^+ ,

$$(5) \quad d^+ = \min_{m \in M^+} \sqrt{E(y_t - m_t)^2},$$

Again, if the model correctly prices every asset, and it does not allow arbitrage opportunities, we should have $d^+ = 0$.

We choose the HJ-distance approach because of the following two advantages. First, the SDF framework is a unified framework for asset pricing analysis. With different risk specifications, different asset pricing models assume different “correct” SDFs. The HJ-distance can be directly used to examine and compare performances of different risk specifications, and then evaluate the pricing errors of the base assets (hedge funds, in our case) with alternative models. Therefore, this approach is much more general and flexible than time series regressions. Second and most importantly, the SDF framework and HJ-distance approach explicitly take into account the arbitrage-free requirement for “correct” SDFs. That is, we require that the SDF of a true asset-pricing model be positive to avoid arbitrage opportunities and ensure correct pricing of derivatives and other nonlinear payoffs. This aspect of the HJ-distance approach is especially appropriate for hedge fund evaluation because hedge funds often rely heavily on derivatives and dynamic trading strategies and, therefore, are capable of “exploiting” negative SDFs. In contrast, it is extremely difficult to impose the arbitrage-free requirement in a linear regression framework.

There are important papers indicating the importance of the arbitrage-free requirement for performance evaluation of actively managed funds. Recent work in the asset pricing literature documents the success of several linear asset-pricing models, and as a result, linear asset pricing models are widely used for performance evaluation applications. However, linear asset pricing models are not arbitrage free, because they allow the SDF to be negative. The use of the linear models on actively managed funds can be problematic. Grinblatt and Titman (1989) argue that models not satisfying the arbitrage-free requirement give incorrect evaluations of mutual fund managers if they short a call option, thereby mimicking outperformance. Hedge fund returns are well known to display option-like behavior. Hsieh and Fung (1997) point out the relation between hedge fund trading strategies and non-linearity in returns. Furthermore, they study one hedge fund strategy, “trend follower”, in Fung and Hsieh (2001). A trend-following fund rapidly switches direction, and may use leverage or derivatives to ratchet its exposure up further. Fung and Hsieh (2001) find that this strategy’s returns are highly nonlinear and similar to the returns on an option strategy called the “lookback straddle”. Mitchell and Pulvino (2001) examine another style of hedge fund, “risk arbitrage”, which is conducted on merger and acquisition

news. This strategy displays outperformance when evaluated with linear models that allow arbitrage but can be explained by the return on an index put. Given the empirical results on nonlinearity in returns and the dynamic trading strategies used by hedge funds, it seems particularly appropriate to apply the arbitrage-free requirement to hedge fund performance. Without the arbitrage-free requirement, linear asset pricing models can over-fit the data and appear to work well (Wang and Zhang 2004) by taking negative values for the SDF. With the arbitrage-free requirement, we can have a fair evaluation of model risk specification and fund performance. It is notable that most of the previous research has ignored the arbitrage-free requirement that provides more appropriate asset pricing results. In this paper, we directly examine the importance of the arbitrage-free requirement, testing model based on d^+ .

The HJ-distance approach offers statistical inferences from four different perspectives. First, the HJ-distances are mispricing measures, and can be directly compared across models. The best model should have the smallest HJ-distances. An alternative interpretation for HJ-distances is that they are the maximum pricing errors for a portfolio with norm set to one. Second, the HJ-distance offers a specification test. If the model is correct, the HJ-distance measures should be zero. Jagannathan and Wang (1996) derive the asymptotic distribution for d under the null hypothesis of correct model. A similar distribution for d^+ is derived in Li, Xu and Zhang (2004). As documented in Ahn and Gadarowski (2004), the asymptotic test for d tends to over reject the null in small samples. To adjust for the small sample bias, the inference based on d and d^+ in this paper relies on empirical distributions from Monte Carlo simulations, as in Zhang (2004). Third, we estimate parameters for each model by minimizing the HJ-distances, following standard GMM procedures.⁸ Following Hansen and Singleton (1982), we also allow risk prices to fluctuate with conditional information by assuming the risk prices are a linear function of time t information variables, Z_t . More details are offered in the model description subsection. The individual parameter estimates provide information on whether individual risk factors are significantly priced for hedge fund returns. Finally, we also examine the pricing errors for individual fund portfolio. We define the pricing error for each portfolio as

$$(6) \quad g_i = E(yR_i - p),$$

⁸ More details of the estimation procedure for d can be found in Hodrick and Zhang (2001) and for d^+ in Wang and Zhang (2004).

which is the difference between the price assigned by the asset pricing model and the real price. As shown in Hodrick and Zhang (2001), the pricing errors are linear transformations of average return errors, which are the differences between sample average returns and model implied expected returns. Therefore, this pricing error is very similar to the traditional Jensen's alpha. We will use g to examine performance of individual hedge fund portfolios.

To investigate whether the arbitrage-free requirement is relevant and important for hedge fund returns, we examine the difference between statistics estimated for both d and d^+ . If the arbitrage-free requirement is not important, we should find that d and d^+ are similar in magnitude. A big difference indicates that the SDF takes many negative values for estimation of d . A useful statistic is the negativity rate, which measures the probability for the SDF estimated under d to take negative values. If the arbitrage-free requirement does not matter, the negativity rate should be close to zero. We can also examine differences between the p -values of the specification tests of $d = 0$ and $d^+ = 0$. Finally, the risk prices and pricing errors estimated under d and d^+ provide more information as to whether the arbitrage-free requirement makes a difference.

Our focus in this paper is to evaluate hedge fund performance, given risk specifications of asset pricing models. Empirically, we must first specify the relevant risk sources, and then examine whether hedge funds have significant abnormal returns after risk adjustment. Thus, portfolio performance evaluation is a joint test of the asset-pricing model and the hedge fund returns. To ensure that our conclusions are not sensitive to risk specifications, we examine a broad variety of asset pricing models with different risk specifications that we introduce in the next subsection.

B. Asset pricing models

In the current asset pricing literature, there are two broad categories of asset pricing models, those with risk specification and those without risk specification. For the first category, the economist specifies the relevant risks that drive the security price movements. For instance, in the CAPM, the market portfolio return defines the investment opportunity set, and is priced. Models with risk specification are either linear or nonlinear. Numerous empirical studies on cross-sectional asset pricing document the success of linear models, and we will focus on linear models in this paper. One caveat applicable to all models with risk specification is that it is impossible to avoid potential model misspecification. To minimize the sensitivity of our results

to any particular risk specification, we include the most important and successful models from the empirical literature on stock pricing and mutual fund pricing.

The SDFs of all linear factor models have the following form:

$$(7) \quad y_{t+1} = b(Z_t)' F_{t+1},$$

where F_{t+1} is a $k \times 1$ factor vector (including a constant term), and $b(Z_t)$ is a $k \times 1$ coefficient vector which linearly depends on an information variable, Z_t . The conditional models are constructed to allow time-variation in factor prices. We choose the conditional variable to capture business cycle variation. For the unconditional models, we set $Z_t = 1$. For conditional models, we use the yield spread between AAA and BAA corporate bonds from Citibase as the conditional variable.⁹ Wang and Zhang (2004) show that allowing every element of factor prices, b , to depend on conditional information tends to make the parameter estimates unstable. As a result, we only allow for the coefficient of the constant factor (intercept element of b) to vary with Z_t , and restrict other elements of b to be constant.

For the second category of asset pricing models, the SDFs are retrieved directly from a set of basis assets following Hansen and Jagannathan (1991) and Ahn, Conrad and Dittmar (2003). These are known as “empirical pricing kernels”. By definition, empirical pricing kernels are linear functions of the returns of basis assets. The advantage of empirical kernels is that they perfectly price all the basis assets, so they presumably reflect all the relevant risks for the basis assets without explicit risk specification. If the hedge fund returns are determined by the same set of risks as the basis asset returns, the empirical pricing kernels should be able to price hedge fund returns. The shortcoming of this approach is that it cannot identify the underlying sources of risk.

In our study, we include four linear models, each with an unconditional version and a conditional version. The first linear model is the CAPM, the typical benchmark in asset pricing tests. It has two factors in its unconditional SDF, a constant and the excess return on the market portfolio. We use the return on the value-weighted CRSP index in excess of the one-month risk free rate as a proxy for the excess return on the market. The conditional SDF has three factors, with the extra factor to be the conditional variable.

The second linear model is the Fama and French (1993) three-factor model (FF hereafter). We choose this model for its successful performance in cross-sectional stock pricing

⁹ We also tried an alternative conditional variable, industrial production (from DRI). The results are similar.

and mutual fund pricing. The first factor is the market factor as in CAPM. The second factor, SMB (small minus big), is constructed as the difference in returns on small and big stocks, and captures risk related to size. The third factor, HML (high minus low), is constructed as the difference in returns between high and low book-to-market stocks, and captures risk related to the B/M ratio. The SDF of the unconditional FF has four factors, including the constant. The conditional SDF has five factors.

While the above factors are important to capture equity and mutual fund returns, they may not be able to capture the option-like payoffs in hedge fund returns. Therefore, our last two linear models are option-based models. Agarwal and Naik (2004) recognize the importance of the non-linearity that hedge fund strategies can induce in hedge fund returns. To capture the option-like payoffs of many hedge fund strategies, they include index put returns in their tests and find that hedge fund returns can be explained in a conventional linear framework. In addition to the market factor, our third linear model includes the return on a nearest-to-at-the-money put option on the S&P 500 index with 20 to 50 days to expiration. We call the third linear model “PUT”. Its unconditional version has three factors, and the conditional version has four factors.

For our fourth model, we aim to capture the systematic risks in the option market. In addition to the market factor, we consider two more systematic factors for the option market. As Ang, Hodrick, Xing and Zhang (2004) indicate that aggregate volatility affects cross-sectional stock returns, we capture aggregate volatility risk by using the return on nearest-to-the-money S&P 500 index straddles. We also add in a jump factor from the option market, which is calculated as the return on out-of-the-money (“money-ness” of approximately 0.92) S&P 500 index puts that expire in 20 to 50 days. The fourth linear model is denoted “OPT”, with four factors for the unconditional version, and five factors for the conditional version. This risk specification is also consistent with the model proposed by Liu and Pan (2003).

We consider three empirical pricing kernels. The first one, IND, is constructed from ten industry portfolio returns. Previous studies (for example, Ahn, Conrad, and Dittmar 2003) have shown that industry portfolios are important for capturing common stock returns. The industry portfolios are constructed at the end of June of year t based on four-digit SIC code at that time, then returns are computed from July of t to June of $t+1$.¹⁰

¹⁰ See Ken French’s web page, http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Since hedge fund managers have access to various investment vehicles, we consider five “alternative” asset classes, which may not be spanned by the industry portfolios. Their pricing kernel is denoted “ALT”. The first additional asset is the CRSP return on an equal-weighted portfolio of closed-end equity funds, which reflects what some authors describe as shifts in small investor demand or “sentiment” (see Lee, Shleifer, and Thaler 1991). The second additional asset is the total return on the Goldman Sachs Commodity Index from Bloomberg.¹¹ Commodity returns may track futures positions held by certain hedge funds and may reflect a variety of systematic and idiosyncratic economic forces (Bailey and Chan 1993). The third additional asset is the total return on the National Association of Real Estate Investment Trusts (NAREIT) equity REIT index that may capture additional unique elements of hedge fund returns.¹² The fourth additional asset is the total return on the J.P. Morgan “EMBI + Brady Broad” index of emerging market hard currency sovereign bonds from Datastream. Many empirical studies use the credit risk implicit in risky bond returns as a risk factor or conditioning variable in asset pricing tests (Ferson and Harvey 1991, Bailey and Chung 1995). The last additional asset is a trade and exchange rate weighted global money market return.¹³ This factor reflects both the course of effective interest rates around the globe and the value of the dollar and, with our other alternative factors, may help explain hedge fund returns better than conventional factors alone. We also pool the industry portfolios and alternative assets, denoting the pricing kernel combining both classes of assets as “BOTH”.

Panel A of Table 1 reports the weights of each component of the three empirical pricing kernels.¹⁴ IND puts heavy weight on the “non durables” and “other” industry portfolios. ALT puts heavy weight on the “Brady Bond” and “Global Yield” series. For the pooled set of IND and ALF assets, the weights on industries stay approximately the same, and we have relatively large weights on “Closed End Fund” and “Global Yield” series. Panel B of Table 1 displays correlations across risk factors. Noticeably, the at-the-money put return is highly correlated with the “straddle” and “out-of-money” return series. As we find later, the performance of “OPT” is similar to, but slightly better than, the performance of “PUT”.

¹¹ The series captures the payoff from monthly rollovers of short maturity commodity index futures contracts, not merely the change in the futures price.

¹² <http://www.nareit.com/nareitindexes/index-d.cfm>

¹³ Dollar returns on short term yields of Belgium, Canada, France, Germany, Italy, Japan, The Netherlands, Sweden, Switzerland, United Kingdom, and United States are weighted by each country’s share of global trade following Harvey (1995).

¹⁴ Note that the kernels are generated at the end of December and, therefore, we assume the weights are constant.

II. Data on Hedge Fund Returns and Characteristics

A. Summary statistics

The hedge fund data used in our analysis is obtained from TASS.¹⁵ Among all the datasets that have been used in existing hedge fund studies, the TASS database is probably the most comprehensive.¹⁶ It covers over 4,000 funds from November 1977 to September 2003, and classifies them in “live” and “graveyard” categories. “Live” funds are those that are active as of September 2003. Once a fund is considered no longer active, it is transferred to the “graveyard” category. The “graveyard” database did not exist before 1994, thus, funds that became inactive before 1994 were not recorded by TASS. To mitigate the problem of survivorship bias,¹⁷ we include both “live” and “dead” funds and restrict our sample to the period between January 1994 and September 2003, yielding a sample of 4131 funds. The database also provides monthly net-of-fee returns and net asset values, plus cross sectional fund characteristics such as management fees and incentives fees.

Table 2 presents summary statistics on monthly hedge fund returns and fund characteristics for the entire sample, and for the subsets of “live” and “dead” funds. Out of the 4131 hedge funds at the end of the sample, 1671 are “dead” and 2460 are “live”. The average monthly return across all funds is about 0.87% per month. Unreported results show that hedge fund returns exhibit significant positive skewness and excess kurtosis. Unsurprisingly, the returns of “live” funds have higher mean and lower standard deviation than those of “dead” funds. Median fund size is about 22 million dollars, and “live” funds are much larger than “dead” funds. Management fees are typically in excess of one percent per year and the incentive fee is typically between 15 and 20 percent of profits. While we would expect to see a higher fee associated with better performance, fees are known to be inflexible across time and do not differ widely across funds. Unreported results indicate that the only discernible pattern is that higher incentive fees seem to be associated with less volatile returns. Most hedge funds use only moderate levels of leverage, with only a few funds reflecting the stereotype that hedge funds are heavily leveraged.¹⁸

¹⁵ For more details on TASS, see www.tassresearch.com/tass.html.

¹⁶ Other sources of data on hedge funds include AltVest, Hedge Fund Research (HFR), and Managed Account Reports (MAR).

¹⁷ See, for example, Ackermann, McEnally, and Ravenscraft (1999) and Brown, Goetzmann, and Park (1997).

¹⁸ Since hedge funds almost always have a “lock up” period (typically 24 months) during which no additional investments or withdrawals are permitted, we also look at sub samples of “young” (age less than or equal to two

We must note potential biases in our dataset. TASS collects data by surveying individual funds. It is possible that fund managers who want to publicize their fund's performance are more likely to complete and return the TASS survey. Thus, selection bias is a concern and, lacking data on additional funds, we cannot address the problem. Furthermore, fund performance is also self-reported, raising concerns about window-dressing or cheating.¹⁹ However, TASS indicates that fund performance is audited yearly, thereby limiting the extent to which performance may be misreported.

B. Hedge fund portfolios

In most of our empirical analysis, we study the performance of hedge fund portfolios. The portfolio approach helps to remove the noise or idiosyncratic risk of individual hedge fund returns, thereby increasing the precision of parameter estimates and the power of statistical tests. The portfolio approach also significantly reduces the dimensionality of the problem and makes the empirical analysis more manageable.

For a comprehensive and robust analysis of hedge fund performance, we choose several different ways to construct hedge fund portfolios. The first approach is based on the eleven styles reported by TASS to classify hedge fund investment approaches. Brown and Goetzmann (2003) report that about a fifth of hedge fund performance is explained by differences in hedge fund style. The substantial cross-sectional dispersion in the style characteristic will facilitate our empirical tests. A detailed description of each style is given in the appendix. This approach allows us to compare the average performance of different hedge fund investment strategies. In general, most strategies can be classified broadly as either directional or relative strategies. The former tend to take directional bets on a wide range of markets, while the latter tend to take hedged bets on relative price discrepancies. In general, “dedicated short”, “emerging market”, “global macro”, “long/short equity” and “managed futures” represent directional strategies, while “convertible arbitrage”, “equity market neutral”, “fixed-income arbitrage” and “event-driven” represent relative strategies. “Fund of funds” and “multiple strategies” reflect a variety of managers or strategies.

years), and “old” (age greater than two years) hedge funds (results not shown here). We find that the distinctions between “old” and “young” funds are not very significant.

¹⁹ Hedge fund returns computed from net asset values may also suffer all the problems inherent in marked-to-market estimates of value, particularly given that many funds may hold unusual, illiquid assets or may be less subject to the “market pressure” of withdrawals and further investments if they are new funds that are locked up or old funds that are closed. See Getmansky, Lo, and Makarov 2004 and Kazemi and Schneeweis 2004.

Panel A of Table 3 reports returns and characteristics of the eleven style portfolios. The value-weighted mean returns of most style portfolios are close to 1%, with “long/short equity” and “global macro” the highest (1.18%) and “dedicated short bias” the lowest (0.53%). “Long/short equity” has the highest average number (over the whole sample) of funds (563), followed by “fund of funds” (369), “managed futures” (211) and “event driven” (178). “Dedicated short” is definitely a minority in the hedge fund industry with an average of only 17 funds in the portfolio. The returns on directional funds tend to be more volatile, measured by the standard deviation of portfolio returns (column four) and the average of volatility of individual funds in the portfolio (column seven), than those of relative funds. Directional funds tend to have higher exposure to the market: “dedicated short” has a beta of about -1, while “emerging market” and “long/short equity” display betas that are close to 0.50. The betas of most other styles are close to zero.²⁰ “Multi-strategy” are typically the largest (average size of \$262 million), followed by “global macro” (average size about \$145 million) and “fixed income arbitrage” (average size about \$118 million). Most funds have a management fee less than 1.5%, but “managed futures” has the highest management fee, 2.4%. “Fund of funds” has the lowest incentive fee of 9.8%, while most other funds have incentive fees between 15% and 20%. “Fixed-income arbitrage” funds have the highest leverage ratio of 3.6, followed by “convertible arbitrage” with 1.48, and “global macro” with 1.23. Most other funds only use moderate leverage.

The correlation matrix of returns on style portfolios in Panel B shows that most relative funds have low correlation with other funds, while directional funds tend to be highly correlated with each other. For example, “long/short equity” is highly negatively correlated with “dedicated short” and positively correlated with “emerging markets” and “event driven.” This suggests that “long/short equity” and “event driven” must have had net long exposure to the market during our sample period, while “emerging markets” had a positive correlation with the U.S. market. “Fund of funds” and “multi-strategy” are highly correlated with each other, and with “long/short equity”.

In addition to style portfolios, we also analyze hedge fund performance based on some commonly used groupings from the asset pricing literature. This serves several purposes. First, alternative portfolio formations allow us to study hedge fund performance from different

²⁰ Given that most common stocks have betas between 0.5 and 1.5, hedge funds obviously provide investment vehicles with very unusual risk properties.

perspectives, thereby yielding more complete and robust results. Second, Brown and Goetzmann (2003) point out that fund manager survey responses are subject to “strategic self-misclassification”.²¹ Therefore, alternative classification schemes may help mitigate the potential problem of strategic self-reporting. Panels C to G report summary information for portfolios sorted on characteristics. Unreported results also examine associations between portfolios sorted by characteristics versus styles.

Panel C reports summary information for portfolios sorted on the market model beta of each hedge fund, which measures the fund’s exposure to the market factor. For each period of 24 months, beta is individually measured for each fund and, at the end of the month, funds are sorted into 10 portfolios according to beta. Thus, rebalancing of the portfolios occurs monthly. Low versus high beta funds display different average returns, but the t-value indicates that this difference is not statistically significant. Unreported statistics indicate that, except for “dedicated short” and “managed futures” that have a heavy concentration in lowest beta (with an average of -1) portfolios and “emerging market” and “long/short equity” with a relatively heavy concentration in the two highest beta portfolios, most styles have a relatively even distribution across different beta portfolios. While the beta portfolios allow us evaluate hedge funds controlling their systematic exposure to the market portfolio, we will not use beta-sorted portfolios in subsequent asset pricing tests because there is not enough return spread across different portfolios to be priced.

Panel D presents results for portfolios sorted by total volatility of returns. Individual firm volatility has proven interesting and important in studies of common stock returns.²² Volatility portfolios are formed in a manner similar to the beta portfolios based on return volatilities estimated using rolling 24-month windows. The return spread between “low volatility” and “high volatility” portfolios is about 1.17% per month, and is statistically significant. The volatility of the “high volatility” portfolio is many times larger than that of the “low volatility” portfolio. Higher volatility tends to be associated with higher beta, smaller size, larger incentive fees, and lower leverage. Consistent with the eleven styles, unreported statistics indicate that there is a

²¹ Given the strategic behavior in style reporting documented by Brown and Goetzmann (2001) for the heavily regulated mutual fund industry, such behavior might be even more severe for the far less regulated hedge fund industry. Fung and Hsieh (1997) argue that “various industry sources frequently publish a much wider range of ‘style classifications.’ Often, reported returns for the same style category will differ across sources and the same manager can appear in different style categories depending on the source.”

²² See, for example, Ang, Hodrick, Xing, and Zhang (2004).

relatively heavy concentration of relative (directional) funds in low (high) volatility portfolios. Volatility portfolios allow us to evaluate hedge funds controlling for total risk.

Panel E presents results for portfolios formed on size, which can be an important factor in determining hedge fund returns. Due to economies of scale, large hedge funds might enjoy lower operation and transactions costs. On the other hand, large funds might be more difficult to manage than small funds and experience worse performance. Each month, hedge funds are sorted into portfolios based on each fund's net asset value six months ago. A size effect is evident: Small funds outperform big funds by a statistically significant difference of about 34 basis points per month. There are some differences in characteristics across size: larger funds tend to have lower fees, more leverage, and are older. Most styles have a relatively even distribution across all size portfolios, except that "emerging market" has a relatively heavy concentration in small size portfolios, and "event driven" and "fixed-income arbitrage" have a relatively heavy concentration in large size portfolios.

We also classify hedge funds based on momentum, that is, whether funds have consistent good or bad performance over a twelve-month horizon. Panel F presents results for portfolios formed on the "6, 6" momentum strategy defined by Jegadeesh and Titman (1993). For each month, hedge funds are sorted into portfolios based on their previous six months' performance. In turn, these portfolios are held for the subsequent six months. It is clear that past winners outperform past losers by a statistically significant difference of about 1.06% per month. Winners also tend to be larger and slightly younger than other funds. Directional funds, perhaps due to their particularly risky strategies, tend to display more extreme performances. About 20% to 25% of these funds fall into both the loser and winner portfolios. On the other hand, unreported statistics indicate that only a small fraction of non-directional funds fall into winner and loser portfolios, with most of them sorted into intermediate portfolios. Momentum portfolios allow us to evaluate funds that have done particularly well or poorly in the recent past.

Our last classification is based on the clustering analysis of Ahn, Conrad, and Dittmar (2003). Unlike previous portfolios that are formed on past information and are investible, the cluster portfolios described in Panel G are formed at the end of our sample period, September 2003, based on the historical return correlation matrix. Thus, this set of assets does not represent a feasible investment strategy. We choose this set of portfolio returns because they maximize dispersion in the payoff space, and they may capture different risk dimensions of the hedge fund

payoff space. The results indicate that the portfolios differ substantially in terms of returns, with portfolio one (ten) having the lowest (highest) mean return. One interesting (unreported) finding is that, across almost all styles, most of the funds (between 40% to 60%) belong to the first two cluster portfolios that have the lowest returns.²³ This suggests that the bottom half of the funds in each style have very similar returns, and the rest of the funds have better performance to varying degrees. Cluster portfolios allow us to examine hedge funds whose historical returns are highly correlated with each other.

Overall, our classification schemes provide different perspectives to dissect hedge fund returns and examine their performance. This information may be particularly interesting to investors in funds of funds and to large institutional investors that construct portfolios of hedge funds.²⁴

III. Empirical Results

A. The HJ-distance without arbitrage-free requirement

Summary statistics

Results on performance evaluation based on the first HJ-distance, d , are presented in the five panels of Table 4. The first row of each panel presents the estimates of the HJ-distances, which can be directly compared across models. Due to the poor small sample performance of the HJ-distance estimates and specification test (Ahn and Gadarowski 2004), we conduct Monte Carlo simulations and use empirical critical values in our statistical inference on model performance. Hence, the second row of each panel presents the small-sample adjusted annual pricing error, for a portfolio with 20% annual volatility. The third row reports the probability that the model can take a negative value, which indicates whether the arbitrage-free requirement is relevant and important. The last row presents the specification test, the small-sample adjusted p-value for the hypothesis $d = 0$.

The first panel presents evaluation results for style portfolios. Overall, the HJ-distances of most models are between 1.2 and 1.4, with the smallest HJ-distance obtained for the conditional OPT, and the largest HJ-distance observed for IND. The magnitude of the pricing errors ranges from 12% for PUT to 20% for CAPM and empirical kernels. Thus, the option factors seem to be

²³ Another highly concentrated portfolio is the eighth cluster portfolio, which has about 50% and 30% of “dedicated short” and “managed futures” respectively. Not surprisingly, that portfolio has a beta that is slightly negative.

²⁴ In addition to what is reported in Table 2, we also examine returns on “live”, “dead”, “young”, and “old” subsets of hedge funds and obtain similar results. Results are not reported but are available on request.

relevant for hedge fund portfolios, while the basis assets cannot span the hedge fund payoff space. The negativity rate of y ranges from 0% for CAPM to 17% for option-related models, which indicates that the arbitrage-free requirement will be important for these option-related models. The specification tests based on d reject every model. Overall, even without the arbitrage-free requirement, the style portfolios are relatively hard to price for all models, although option models seem to be able to capture some relevant risks.

The results in the second panel show that volatility portfolios are somewhat easier to price than style portfolios. The HJ-distances are all between 0.808 and 1.007, and corresponding errors are between 7% and 10% for a portfolio with 20% annual volatility. The conditional FF model has the smallest HJ-distances. Again, IND has the largest pricing error. From the negativity rate, the multi-factor models and conditional models take more negative values. For the specification test, small p-values indicate that all models are rejected at the 5% level (while a couple pass at the 1% level), particularly the unconditional models and empirical kernels.

The next three panels show that the evaluation results for size, momentum, and cluster portfolios are qualitatively similar to each other. In terms of the magnitude of HJ-distances, size, momentum and cluster portfolios are easier to price than style and volatility portfolios. The maximum pricing errors in the second row of each panel are at most 6% per year.²⁵ Consistently, we cannot reject the majority of the models. Either the conditional FF or the conditional OPT has the best performance in explaining the returns of these three characteristic-sorted portfolios. The worst models are either CAPM or empirical kernels, and they have similar performances. It is possible that, since IND is derived from industry returns, it mainly captures market risk as in the CAPM. However, the risks in both models are not adequate for pricing hedge fund returns. It is also evident that conditional models improve HJ-distances significantly. The negativity rate is high for multi-factor and conditional models. The specification test indicates conditional FF and conditional OPT are able to price these characteristics-sorted portfolios.

To summarize, the evaluation results based on the first HJ-distance measure, d , without arbitrage-free requirement suggest that the returns on hedge fund portfolios sorted by characteristics can be explained by several linear asset pricing models, especially the option-

²⁵ A 0% pricing error means that the estimated HJ-distance is smaller than the small sample average HJ-distance under the null hypothesis that the model is correct.

based models, Fama-French models, and models with time varying risk prices. However, all the models are rejected when pricing the style and volatility portfolios.

Risk prices

Factor price estimates are defined as in equation (7). The factor prices provide information on whether individual risk factors are relevant and important for the assets we want to price. The parameters and their standard errors are calculated as in Hodrick and Zhang (2001). Table 5 reports the statistics. For each panel, the first line contains parameter estimates and the second line contains the corresponding t-tests. We only report statistics for the linear asset pricing models because there are no risk prices for the empirical kernels.

Panel A provides factor prices for style portfolios. Out of the eight risk factors (from both stock market and option market), the market factor and two factors from the option market, PUT and STR, are significantly priced. The conditioning variable is marginally significant at the 10% level for the stock market models. In a nutshell, the option market factors seem to be relevant for pricing the style portfolios.

Panels B through E provides factor prices for characteristic-sorted portfolios. For the volatility portfolios in Panel B, the market factor is significant or marginally significant for the first two models, the STR option-related factor is marginally significant, and the conditioning variable is marginally significant for most cases. This implies that the volatility portfolio returns are related to market volatility. For size portfolios in Panel C, the market factor and the PUT factor are significant. Interestingly, the size and B/M factors from the stock market are both marginally significant. This implies that size portfolios from hedge fund space have similar driving forces as size portfolios from the stock market. For the momentum portfolios in Panel D, the conditioning variable is consistently significant. PUT is also significant in one case. For the cluster portfolios in Panel E, the size factor is significant and the B/M factors from the stock market and the PUT and STR factors from the option market are close to marginally significant. This indicates that both sets of factors are relevant for at least some subgroups of hedge funds.

Pricing errors

Table 6 provides the pricing errors as defined in equation (6). This allows us to examine individual portfolio errors. Bold entries represent pricing errors that are significant at the 5% level. Panel A focuses on the pricing errors of the eleven style portfolios under different models. Consistent with their overall performances measured by d , the CAPM and FF have significantly

positive pricing errors for most funds. IND, ALT, and BOTH have slightly better performance, with fewer funds displaying significant pricing errors. The pricing errors for most of the eleven style portfolios are insignificant under the option-related models. In fact, a couple of funds have negative (although not significant) pricing errors under the conditional OPT. However, the model errors under d can be misleading: If the arbitrage-free requirement is not imposed, the models can over fit the data.

Panel B reports the pricing errors of volatility portfolios. In general, the magnitude of the pricing errors is the highest for the low volatility portfolios. The pricing errors of the conditional FF are not significant and not even uniformly positive. This again indicates that it is easier to capture the returns of the volatility portfolios than the style portfolios.

The pricing errors reported in Panel C for size portfolios show that small funds tend to have higher pricing errors than large funds. Under most models, the pricing errors of the smallest size portfolio are about two times or more of the size of those of the largest size portfolio. While most models, especially the CAPM and the empirical kernels, generate significant positive pricing errors for most size portfolios, there is no evidence of superior performance after risk-adjustment using FF or option-related models.

Evaluation results for momentum portfolios in Panel D show that, under most models (except conditional FF), the loser portfolio has negative pricing errors while the winner portfolio has positive pricing errors. This indicates that most models overprice losers and underprice winners. The evidence of abnormal performance and the pattern in pricing errors of loser and winner portfolios disappear after risk adjustment using the conditional FF model.

The results for cluster portfolios in Panel E are similar to that of other characteristic portfolios. Both FF and option models are better than the other models at explaining the returns of cluster portfolios.

Table 6 reveals several general patterns in hedge fund performance measured by the HJ-distance without arbitrage-free requirement. First, the CAPM and empirical kernels are not very successful in capturing hedge fund returns and significantly underperform other models. Second, most models indicate that most hedge fund portfolios have positive pricing errors, with high volatility, small size, and winner momentum portfolios having higher pricing errors than low volatility, big size, and loser momentum portfolios respectively. Third, the two option models and the Fama-French models help to price the style, volatility, size, and cluster portfolios. This

implies important factors in the stock and option markets are also relevant for the hedge fund market. However, the style portfolios and volatility portfolios are much harder to price.

B. The HJ-distance with arbitrage-free requirement

One of the main concerns with the above results is that the SDFs of some models can take negative values and assign inappropriate prices to derivatives and dynamic trading strategies. This is especially true for models with highly volatile factors, such as the option models and conditional models. To address this concern, we repeat the above analyses, explicitly restricting the SDFs of the correct model to be positive to avoid arbitrage opportunities.²⁶

Summary statistics

The results of performance evaluation based on d^+ are reported in Table 7. The first row reports the estimated HJ-distance with the arbitrage-free requirement, d^+ . When compared to d in Table 4, big differences between d and d^+ indicate that the SDFs of these models can take nontrivial negative values without the arbitrage-free requirement. The second row presents the corresponding pricing error for a portfolio with 20% annual volatility for each d^+ measure, after adjustment for small-sample bias. The third row reports the probability the pricing kernel takes a negative value with the arbitrage-free requirement.²⁷ The last row reports the empirical p-value of the hypothesis $d^+ = 0$ after adjusting for small sample bias.

The first panel of Table 7 reports results for style portfolios. For all models, d^+ is larger, by about 80% or more, than d reported in Table 4. This suggests that the performance of most models is inflated by allowing the SDFs to be negative, thus implying arbitrage opportunities. This is especially true for conditional models, because the conditioning variable is very volatile itself. With the arbitrage-free requirement, the pricing errors are between 32% and 37% per year. For the negativity rate, with the arbitrage-free requirement, the models at most take negative values 6% of the time. This implies the arbitrage-free requirement is effective in reducing the chance that linear models take negative values to fit the data better. Consistent with the huge errors, none of the models can pass the specification test, $p(d^+=0)$. The best performing models are conditional OPT and conditional PUT, while the worst is IND.

The evaluation results for volatility portfolios in the second panel of Table 7 show that d^+ is about 10-20% larger than d reported in Table 4, and the annual pricing error for the portfolio

²⁶ This is consistent with findings in Wang and Zhang (2004).

²⁷ Since the models are linear, it is always possible that $p(y<0)$ is positive. When we restrict the SDF to be positive, we are restricting the corresponding m to be positive, rather than truncating the SDF y implied by the model.

with 20% annual volatility is about 1-3% larger. Using d as in Table 4, some of the models earn a p-value on the specification test that is greater than 1% significance. However, using d^+ , all models fail the specification test for volatility portfolios at any level of significance. While the conditional option model has the smallest d , the conditional FF has the smallest d^+ , suggesting that the seemingly superior performance of option models on volatility portfolios is due to the fact that the volatile option returns allow the SDF to take negative values.

Across the next three panels of Table 7, the results for size, momentum, and cluster portfolios are similar. For most models, d^+ is about 10-20% bigger than d , although for momentum and cluster portfolios, d^+ typically exceeds d in conditional models even more noticeably. Either FF or option models typically have the best performance in capturing the returns of the three characteristic portfolios and cannot be rejected by the specification tests.

Comparing the results in Table 4 and Table 7, we find that the arbitrage-free requirement makes a difference from three perspectives. First, the HJ-distance magnitude increases substantially, especially for style portfolios and volatility portfolios. Since arbitrage is one part of the pricing error measured by the second HJ-distance, but not by the first HJ-distance, this indicates that overlooking the arbitrage opportunities yields better data-fitting and seemingly smaller HJ-distances. The problem is especially relevant for style portfolios. Second, since the magnitude of HJ-distances increases with the arbitrage-free requirement, it is much harder to pass the specification test. We still cannot price the style and volatility portfolios, while for the other portfolios, the p-values drop from around 80% to around 10%. Finally, from the negativity rate, imposing the arbitrage-free requirement effectively cuts down the chance that the SDF taking negative values. For instance, for style portfolios, the negativity rate decreases from 17% (PUT for d) to 4% (PUT for d^+).

Factor prices

The factor prices under the arbitrage-free requirement are presented in Table 8. The asymptotic distributions of parameters are derived in Li, Xu and Zhang (2004). For style portfolios in Panel A, most of the parameters are noticeably smaller than those in Panel A of Table 5. This makes sense because, with the arbitrage-free requirement, the parameters have less freedom to take wild swings to fit the data. In terms of significance, we find that the option factors, PUT and STR, are significantly priced for the style portfolios, as is MKT. For the volatility portfolios in Panel B, the stock market factors SMB and HML gain some significance,

while the option market factors, PUT and STR, lose significance. For other characteristics-sorted portfolios in Panels C to E, the results are similar to those in Table 5, except the magnitude of parameters is smaller and the stock market factors sometimes gain more significance.

Comparing Table 8 and Table 5, we can examine the differences in parameter estimates with and without the arbitrage-free requirement. First, the parameters are smaller with the arbitrage-free requirement. A time-series plot of the estimated SDF (not reported but available on request) demonstrates that smaller parameters restrict the chance that the SDF takes negative values, while bigger parameters can yield extreme and negative values that fit the data better. This is consistent with the negativity rate finding. Second, the stock market factors are more important with the arbitrage-free requirement, while the conditioning variable loses significance. This is understandable because the conditioning variable is highly volatile, thereby allowing the SDF to take negative and extreme values. Thus, the conditioning variable appears important for the estimates without the arbitrage-free requirement. The opposite rationale works for the stock market factors.

Pricing errors

Table 9 reports the pricing errors for individual hedge fund portfolios estimated under d^+ . A comparison between Table 9 and Table 6 (under d) suggests that the arbitrage-free requirement yields material differences in cross-sectional errors. Without the arbitrage-free requirement, the option models come close to explaining most of the returns of all style portfolios (Panel A of Table 6). With the arbitrage-free requirement, none of the models looks particularly good at capturing the returns of the style portfolios. In particular, “convertible arbitrage”, “equity market neutral” and “long/short equity hedge” have significantly positive pricing errors under d^+ for almost all models.

The other panels of Table 9 show that the impact of the arbitrage-free requirement on the evaluation results of volatility, size, momentum, and cluster portfolios is not as significant as that on style portfolios. The pricing errors under d^+ are generally bigger than those under d . While certain models can capture the returns of some characteristic portfolios under d , they fail to do so under d^+ . For example, we see frequent significant positive pricing errors for volatility and size portfolios. While the arbitrage-free requirement increases the magnitude of the pricing errors of momentum and clustering portfolios, the pricing errors under most models for most portfolios remain insignificant.

To summarize, the results in Tables 7 to 9 show that the arbitrage-free requirement significantly increases the magnitudes of HJ-distances, decreases the magnitude of parameter estimates, and increases pricing errors for most hedge fund portfolios, especially when using conditional models and multi-factor models. As a result, no model can adequately explain the returns of style portfolios and volatility-sorted portfolios, although either FF or OPT can typically explain the returns of other characteristic portfolios. In particular, three styles, “convertible arbitrage”, “equity market neutral” and “long/short equity hedge”, have significant positive pricing errors under all models. These results indicate that imposing the arbitrage-free requirement is essential for understanding hedge fund performance more completely.²⁸ Our asset pricing results clearly indicate a need for better models of hedge fund returns.

C. What is missing from the models for style portfolios?

Since none of our models adequately price the style portfolios, we take a closer look at what we can possibly be missing by using common models inspired by conventional studies of the stock market and other financial assets. We apply an empirical kernel approach which lacks risk specification but, by definition, prices hedge fund returns perfectly.

Panel A of Table 10 reports the weights used to construct a “style empirical kernel” from the individual style portfolios. The highest weight (68.3%) is put on “equity market neutral”, which indicates this particular style reflects much of the risk in hedge fund payoff space. As we learned from our previous results, none of the models, especially with the arbitrage-free requirement imposed, can price this particular style. The second largest weight (32.2%) is on “event-driven”. The third largest weight (17.6%) is on “long/short equity hedge”, which is also problematic for all models. Interestingly, “fund of funds” has a large negative weight (-56.0%), which implies that this style might underperform other styles and, therefore, is shorted in hedge fund space.

Panel B of Table 10 reports correlations between this “style empirical kernel” and our models. Bold fonts indicate significance at 5% level. All of our models have correlation lower than 25% with the style empirical kernel. The highest (22%) is observed for option-based models. Among the models, we find relatively high correlations within groups, for instance, CAPM versus FF, or among option related models. This demonstrates that all of our models

²⁸ We check for robustness by excluding influential observations including the LTCM crash period. Our results are unchanged.

ignore features (risks or characteristics) that are important for assessing hedge fund performance and, therefore cannot pass the specification test.

There are two caveats regarding the use of the style empirical kernel. First, it does not identify the driving forces behind hedge fund pricing. The kernel, by itself, does not reveal if additional market wide risks, space-specific risks, or fund characteristics (like managerial skill) are most critical. Since we have used most of the common models and concepts from the stock pricing literature, we explore one potential avenue, associations between manager’s skills and apparent mispricing, in the next subsection. Second, the style empirical kernel is not constrained to be arbitrage free if we want “perfect” pricing. To be consistent with the scheme of this paper, we truncate the style empirical kernel by setting its value to zero whenever it takes a negative value. We check the pricing error for this truncated kernel, and the pricing errors for the individual style portfolios that comprise the kernel are still small and insignificant.

D. Pricing individual funds with fund and manager characteristics

In previous subsections, we conducted pricing tests at the portfolio level. The advantage of portfolio level pricing is that it reduces the idiosyncratic noise from the fund level. Since we find that none of our models is able to pass the specification test for style portfolios, one alternative is to look at fund characteristics and fund manager characteristics. However, this is not feasible at the portfolio level. In this subsection, therefore, we directly examine individual fund performance using the risk adjustment from asset pricing models, fund characteristics, and fund manager characteristics. This complements our study at the portfolio level. Meanwhile, by using fund characteristics and fund manager characteristics, we seek to understand whether the apparent mispricing of our stock market and option market models when applied to style portfolios is related to fund manager skills. We use the manager’s experience, age, and quality of education as proxies for fund manager skill. If characteristics help to explain the abnormal performance, it will be useful to condition predictions or ex-post tests of performance on these characteristics as part of the process of hedge fund evaluation and selection.

To risk adjust individual fund returns, we take our models estimated at the portfolio level literally, without re-estimating parameters. Most of the previous literature re-estimates the parameters for individual funds, which probably fits the data better. However, our purpose is to keep the asset-pricing model (and risk adjustment) consistent throughout the paper. Therefore, we choose the best model, OPT, from the d^+ estimation. Since this model still takes negative

value with 4% probability, we truncate it when the SDF takes negative value. The distance estimate is slightly larger for the truncated model. We know this model cannot perfectly adjust hedge fund returns for risk, but we take it as the best choice from our models inspired by the stock pricing and mutual fund pricing literature. For comparison, we also include the style empirical kernel just described as an alternative for risk adjustment.

Related research on fund manager characteristics is typically conducted on mutual fund data. Chevalier and Ellison (1999a), for example, test whether the quality of a mutual fund manager's education (average SAT score of undergraduate institution, whether or not the manager holds an MBA degree) explain mutual fund performance. After adjusting for risk using a four-factor Fama and French model on individual fund returns and controlling for other managerial and fund characteristics, mutual fund performance is positively correlated with the average SAT score of the manager's undergraduate degree-granting institution. Performance is also negatively correlated with the age of the manager. The more detailed examination of managerial incentives in Chevalier and Ellison (1999b) describes additional associations between manager age, performance, and risk-taking.²⁹

We collect data for fund managers from the TASS "notes" files. Table 11 provides summary statistics for the relevant data items. We only collect data for the primary manager in each fund. For the SAT score, we first collect the undergraduate school for each fund's primary manager, then collect the mean SAT score for admission from the latest *Princeton Review* and *USA Today* data. For the manager's birth year, if it is not indicated, we use the graduation year from the undergraduate program minus 22 years. Manager's tenure starting year is given in the notes file. Manager's experience starting year, if not given in the dataset, is estimated by the MBA graduation year. The dummy variable "MBA" has a value of 1 if the manager has an MBA degree, and 0 otherwise. The dummy variable "CFA" has a value of 1 if the manager passes the CFA exam, and 0 otherwise.

From Panel A of Table 11, we can see that we have many missing values, especially for fund manager's age. Compared, however, to Chevalier and Ellison (1997a), we have a

²⁹ Recent studies of individual or household portfolios and characteristics also yield interesting evidence of associations between portfolio performance and characteristics of the portfolio manager. For example, Barber and Odean (2001) compare the performance of stock portfolios held by male versus female individual investors. They find that men are more aggressive, turn over their portfolios more frequently, and experience poorer performance relative to women. However, our sample has only around 80 female managers out of 4000 an insufficient sample for inference about the impact of gender difference.

reasonable sample size. The average SAT score of the manager's undergraduate school ranges from 865 to 1520, with a mean of 1305. On average, managers started at their fund in 1996, started in the business over ten years earlier, and were born in 1960. The lead manager is typically identified with other managers. In a minority of cases, he or she holds an MBA or CFA and is a graduate of an Ivy League college.

Panel B of Table 11 presents the correlations between raw returns, pricing errors (after risk adjustment), and fund and manager characteristics. Pricing errors are defined in equation (6). Each correlation is constructed with the maximum number of monthly returns available for each fund. The correlations are equivalent to univariate regressions, where the squared correlation is the R-squared coefficient and significance equals regression coefficient significance. The first column of the correlation matrix studies raw returns. We find that raw returns are significantly and positively correlated with SAT score, dummy for Ivy League undergraduate education, and dummy for holding an MBA degree. This suggests that individual fund returns are related to the education and quality of their manager. The raw returns are significantly and negatively correlated with fund manager tenure, which is consistent with the finding by previous authors that managers become more conservative over time. Finally, raw returns are also positively and significantly associated with incentive fee and management fee. The second and third columns cover risk-adjusted returns generated by conditional OPT model and the style empirical kernel respectively. The results are similar to those reported for raw returns, but with smaller magnitudes, especially for the kernel adjusted returns. This indicates that part of the correlation is related to the risk adjustment, and it disappears after risk adjustment. Overall, the univariate results show that there are a number of associations between fund and fund manager characteristics and fund returns.

Next, we estimate several simple multivariate regressions of the raw return and pricing errors on the characteristics. We have two concerns about the regression specifications. First, from the correlation matrix, we find that IVY dummy and SAT score are highly correlated (70%). Since we do not know, *ex ante*, which is more relevant, we will have one set of regressions using SAT and another using the IVY dummy. Second, we also find relatively high correlation (although lower than the IVY and SAT correlation) between manager's age, manager's experience, and manager's tenure. Therefore, we examine several specifications using

combinations of those characteristics. We report two variations, one using all three variables and one using only experience and tenure. Other specifications give similar results.

The regression results are presented in Table 12. For raw returns, we find the SAT score is marginally significant. Manager's age and manager's tenure are significant, while the fund characteristics are not significant. For the OPT errors, we find marginal significance for SAT and manager's age, and significance for manager's tenure. Notice that the intercept term is smaller and less significant than that of the raw returns, which indicates that risk adjustment takes away part of the explanatory power of the characteristics. For the STYLE errors, it is intriguing to find that none of the characteristics (except the manager's tenure) matter for the error. This implies that the empirical kernel already takes into account the characteristics of the funds and their managers.

Since our focus is on explicit risk adjustment models, we take a closer look of the regression results for the OPT error. We can consider the economic significance of the estimated regression slopes as follows. The coefficient on SAT is typically around 0.001%, indicating that a fund manager from an undergraduate school with a 100-point higher average SAT score typically produces an extra 1.2% per year. The coefficient on manager's age is between -0.011% and -0.017%. For every five years added to manager's age, return declines by about 0.66% to 1.02% per year. The coefficient on manager's tenure is between -0.016% and -0.029%. For every five years added to manager's tenure, return declines by about 0.96% to 1.74% per year. When we exclude the manager's age, the significance and impact move to the manager's experience variable. The results are similar to those of manager's age and manager's tenure.

When the dummy for IVY replaces the SAT score as a proxy for the quality of the manager's education, it is always significant for the raw return and the OPT error, though it is not significant for the STYLE error. Comparing the intercept terms of SAT and IVY specifications, we find that the SAT score is more helpful in explaining the returns and errors than the IVY dummy, which has less information in cross-section.

Our earlier results on portfolios demonstrate that different hedge fund styles yield different patterns in returns that our models have trouble fully explaining. Pooling results across all styles, as in Table 12, may be too restrictive. Therefore, we repeat the regressions of individually for three styles that we had particular trouble pricing earlier. They are

“convertible arbitrage”, “equity market neutral” and “long/short equity hedge”. The results (not reported but available on request) are qualitatively similar. We find that the manager’s characteristics help to explain OPT errors, while they have no impact on the STYLE errors. We also find the intercepts for these three individual styles are not significant.

Overall, the regressions of performance on characteristics indicate that the fund manager’s education and experience help explain what is left of returns after explicit risk adjustment. However, those characteristics are not important for the empirical kernel pricing errors.

IV. Conclusion

In this paper, we perform a comprehensive evaluation of hedge fund performance using the stochastic discount factor (SDF) approach. What differentiates our research from previous studies is that we use a general SDF and HJ-distance approach, and explicitly take into account the arbitrage-free requirement on asset pricing models. As a result, our approach ensures appropriate valuation of derivatives and dynamic trading strategies. This is especially important for evaluating the performance of hedge funds that are often heavily involved in these practices. In contrast, it is much more difficult to impose the same restriction in a traditional linear regression framework, and results on hedge fund performance based on Jensen’s alpha can be difficult to interpret.

Using the SDFs of a wide variety of asset pricing models, we evaluate the performance of hedge fund portfolios sorted by characteristics and styles. There are three asset-pricing findings. First, we find significant return spreads among hedge fund portfolios sorted on various characteristics. Second, without the arbitrage-free requirement, most linear asset pricing models are able to explain the returns on characteristic sorted portfolios, but not the style portfolios. Third, with the arbitrage-free requirement, it is much more difficult to price the hedge fund portfolios, which indicates the importance of the arbitrage free requirement. Some models are still able to price the size, momentum and cluster portfolios, but none of the models can possibly price style portfolios and volatility portfolios.

The results on spreads should be helpful for the asset allocation decisions of “funds of funds”, for hedge fund investors generally, and for hedge fund managers themselves. The spreads indicate that performance clearly varies across styles and other easily measured attributes. Perhaps more importantly, the asset pricing results yield a stark conclusion. Either

existing models based on stock and option market factors cannot adequately adjust hedge fund returns for risk or certain types of hedge funds have typically delivered significant outperformance.

We explore potential deficiencies in our array of asset pricing models by constructing an empirical pricing kernel from the style portfolios. The correlation between the models with stock or option factors and the empirical kernel from hedge funds again illustrates that models derived from stock and option factors miss information that is relevant and important to the pricing of hedge funds, even though previous researchers have found such models successful in pricing stocks and mutual funds. This challenges researchers to find better risk specifications to explain hedge fund returns.

In addition to illustrating the impact of missing risks, we also examine the relation between fund and managerial characteristics and risk-adjusted individual fund returns. When hedge fund returns are adjusted using the empirical pricing kernel built from the style portfolios, fund characteristics do not matter. This is sensible as the empirical kernel naturally reflects all the factors that determine hedge fund pricing. If, however, hedge fund returns are adjusted using our best model from the asset-pricing tests, the manager's age, experience, and educational background explain a significant fraction of hedge fund returns. Aside from offering practical insights into the determinants of hedge fund performance, these results echo our finding that hedge funds are difficult to price using conventional models and also suggest the importance of characteristics, in addition to risk factors, for pricing these interesting and important securities.

Appendix. TASS Style Definitions

The following classification of hedge fund categories is used in TASS dataset.

Convertible Arbitrage: This strategy is identified by hedge investing in the convertible securities of a company. A typical investment is to be long the convertible bond and short the common stock of the same company. Positions are designed to generate profits from the fixed income security as well as the short sale of stock, while protecting principal from market moves.

Dedicated Short Bias: Short biased managers take short positions in mostly equities and derivatives. The short bias of a manager's portfolio must be constantly greater than zero to be classified in this category.

Emerging Markets: This strategy involves equity or fixed income investing in emerging markets around the world. As many emerging markets do not allow short selling, futures, or other derivative products with which to hedge, emerging market investing often employs a long-only strategy.

Equity Market Neutral: This investment strategy is designed to exploit equity market inefficiencies and usually involves being simultaneously long and short matched equity portfolios of the same size within a country. Market neutral portfolios are designed to be either beta or currency neutral, or both.

Event Driven: This strategy is defined as 'special situations' investing designed to capture price movement related to a significant pending corporate event such as a merger, corporate restructuring, liquidation, bankruptcy or reorganization. There are three popular sub-categories in event-driven strategies: risk (merger) arbitrage, distressed/high yield securities, and Regulation D.

- **Risk Arbitrage:** Specialists invest simultaneously long and short in the companies involved in a merger or acquisition. Risk arbitrageurs are typically long the stock of the company being acquired and short the stock of the acquirer.
- **Distressed Securities:** Fund managers invest in the debt, equity or trade claims of companies in financial distress or already in default. The securities of companies in distressed or defaulted situations typically trade at substantial discounts to par value and thereby attract investments when managers perceive that a turn-around will materialize.
- **High Yield:** This strategy refers to investing in low-grade fixed-income securities of companies that show significant upside potential. Managers typically buy and hold high yield debts.
- **Regulation D:** This strategy refers to investments in micro and small capitalization public companies that are raising money in private capital markets. Investments are usually convertible securities with an exercise price that floats or a look-back provision that insulates the investor from a decline in the price of the underlying stock.

Fixed Income Arbitrage: The fixed income arbitrageur aims to profit from price anomalies between related interest rate securities. Most managers trade globally with a goal of generating steady returns with low volatility. This category includes interest rate swap arbitrage, US and non-US government bond arbitrage, forward yield curve arbitrage, and mortgage-backed securities arbitrage. The mortgage-backed market is primarily US-based, over-the-counter and particularly complex.

Fund of Funds: A multi-manager fund will employ the services of two or more trading advisors or hedge funds who will be allocated cash by the trading manager to trade on behalf of the fund.

Global Macro: Global macro managers carry long and short positions in any of the world's major capital or derivative markets. These positions reflect their views on overall market direction as influenced by major economic trends and or events. The portfolios of these funds can include stocks, bonds, currencies, and commodities in the form of cash or derivatives instruments. Most funds invest globally in both developed and emerging markets.

Long/Short Equity: This directional strategy involves equity-oriented investing on both the long and short sides of the market. The objective is not to be market neutral. Managers have the ability to shift from value to growth, from small to medium to large capitalization stocks, and from a net long position to a net short position. Managers may use futures and options to hedge. The focus may be regional, such as long/short US or European equity, or sector specific, such as long and short technology or healthcare stocks. Long/short equity funds tend to build and hold portfolios that are substantially more concentrated than those of traditional stock funds.

Managed Futures: This strategy invests in listed financial and commodity futures markets and currency markets around the world. The managers are usually referred to as Commodity Trading Advisors, or CTAs. Trading disciplines are generally systematic or discretionary. Systematic traders tend to use price and market specific information (often technical) to make trading decisions, while discretionary managers use a judgmental approach.

Multi-Strategy: This strategy tends to use more than one strategy and cannot be classified in one of the ten styles.

References

- Ackermann, Carl, McEnally, Richard, and David Ravenscraft, 1999, The Performance of Hedge Funds: Risk, Return, and Incentives, *Journal of Finance* 54, 833-874.
- Agarwal, Vikas, and Narayan Y. Naik, 2004, Risks and Portfolio Decisions Involving Hedge Funds, forthcoming in *Review of Financial Studies*.
- Ahn, Dong-Hyun, Conrad, Jennifer, and Robert F. Dittmar, 2003, Risk Adjustment and Trading Strategies, *Review of Financial Studies* 16, 459-485.
- Ahn, Seung, and Christopher Gadarowski, 2004, Small sample properties of the GMM specification test based on the Hansen-Jagannathan distance, *Journal of Empirical Finance* 11, 109–132.
- Akerlof, George A., 1970, The Markets for “Lemons”: Quality Uncertainty and the Market Mechanism, *Quarterly Journal of Economics* 84, 488-500.
- Ang, Andrew, Hodrick, Robert J., Xing, Yuhang, and Xiaoyan Zhang, 2004, The Cross Section of Volatility and Expected Returns, forthcoming in *Journal of Finance*.
- Bailey, Warren, and K. C. Chan, 1993, Macroeconomic Influences and the Variability of the Commodity Futures Basis, *The Journal of Finance* 48, 555-73.
- Bailey, Warren, and Y. Peter Chung, 1995, Exchange Rate Fluctuations, Political Risk, and Stock Returns: Some Evidence from an Emerging Market, *Journal of Financial and Quantitative Analysis* 30, 541-61.
- Barber, Brad and Terry Odean, 2001, Boys will be Boys: Gender, Overconfidence, and Common Stock Investment, *Quarterly Journal of Economics* 116, 261-292.
- Ben Dor, Arik, and Ravi Jagannathan, 2002, Understanding Mutual Fund and Hedge Fund Styles using Return Based Style Analysis, NBER Working Paper 9111.

- Brown, Stephen J., and William N. Goetzmann, 2003, Hedge Funds with Style, *Journal of Portfolio Management* 29, 101-113.
- Buraschi, Andrea and Jens Jackwerth, 2001, The Price of a Smile: Hedging and Spanning in Option Markets, *Review of Financial Studies* 14, 495-527.
- Chen, Zhiwu. and P. J. Knez, 1996, Portfolio Measurement: Theory and Applications, *Review of Financial Studies* 9, 511–555.
- Chevalier, Judith and Glenn Ellison, 1999a, Are Some Mutual Fund Managers Better than Others? Cross-Sectional Patterns in Behavior and Performance, *Journal of Finance* 54, 875-899.
- Chevalier, Judith and Glenn Ellison, 1999b, Career Concerns of Mutual Fund Managers, *Quarterly Journal of Economics* 114, 389-432.
- Coval, Joshua D., and Tyler Shumway. Expected Option Returns, *Journal of Finance* 56, 983-1009.
- Cochrance, John, 2001, Asset Pricing, Princeton University Press.
- Dybvig, Phil and Jonathan Ingersoll, Mean-Variance Theory in Complete Markets, 1982, *Journal of Business* 55, 233-251.
- Dybvig, Phil and Steve Ross, 1985, The Analytics of Performance Measurement Using a Security Market Line, *Journal of Finance* 40, 401–416.
- Fama, Eugene and Kenneth French, 1993, Common Risk Factors in the Returns on Stocks and Bonds, *Journal of Financial Economics*, 3-56.
- Ferson, Wayne, Tyler Henry and Darren Kisgen, 2003, Evaluating Government Bond Fund Performance with Stochastic Discount Factors, unpublished working paper.
- Ferson, Wayne E., and Campbell R. Harvey, 1991, The Variation of Economic Risk Premiums”, *Journal of Political Economy*, 99, 385-415.

- Fung, William, and David A. Hsieh, 1997, Empirical Characteristics of Dynamic Trading Strategies: The Case of Hedge Funds, *Review of Financial Studies* 10, 275-302.
- Fung, William, and David A. Hsieh, 2001, The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers, *Review of Financial Studies* 14, 313-341.
- Getmansky, Mila, Lo, Andrew W., and Igor Makarov, 2004, An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns, *Journal of Financial Economics*, 74, 529-610.
- Glosten, Larry and Ravi Jagannathan, 1994, A Contingent Claim Approach to Performance Evaluation, *Journal of Empirical Finance*, 133-160.
- Goetzmann, William, Ingersoll, Jonathan, Spiegel, Matthew, and Ivo Welch, 2002, Sharpening Sharpe Ratios, NBER Working Paper 9116.
- Grinblatt, Mark and Sheridan Titman, 1989, Portfolio performance evaluation: Old issues and new insights, *Review of Financial Studies* 2, 393-421.
- Hansen, Lars Peter, and Ravi Jagannathan, 1991, Implications of Security Market Data for Models of Dynamic Economies, *Journal of Political Economy* 99, 225–262.
- Hansen, Lars Peter, and Ravi Jagannathan, 1997, Assessing Specification Errors in Stochastic Discount Factor Models, *Journal of Finance* 52, 557–590.
- Hansen, Lars Peter and Kenneth Singleton, 1982, Generalized Instrumental Variables Estimation of Nonlinear Rational Expectations Models, *Econometrica* 50, 1269-1286.
- Harvey, Campbell R., 1995, Global Risk Exposure to a Trade-Weighted Foreign Currency Index, unpublished Duke University working paper.
- Hodrick, Robert and Xiaoyan Zhang, 2001, Evaluating the Specification Errors of Asset Pricing Models, *Journal of Financial Economics* 62, 327-376.

- Hsieh, David, and William Fung, 1997, Empirical Characteristics of Dynamic Trading Strategies: The Case of Hedge Funds, *Review of Financial Studies* 10, 275-302.
- Jagannathan, Ravi and Zhenyu Wang, 1996, The Conditional CAPM and the Cross-Section of Expected Returns, *Journal of Finance* 51, 3-54.
- Jagannathan, Ravi, and Zhenyu Wang, 2002, Empirical Evaluation of Asset Pricing Models: A Comparison of the SDF and Beta Methods, *Journal of Finance* 57, 2337–2367.
- Jegadeesh, N., and Sheridan Titman, 1993, Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency, *Journal of Finance* 48, 65-91.
- Kazemi, Hossein, and Thomas Schneeweis, 2004, Hedge Funds: Stale Prices Revisited, unpublished University of Massachusetts Amherst working paper.
- Lee, Charles, Shleifer, Andre, and Richard Thaler, 1991, Investor Sentiment and the Closed End Fund Puzzle”, *Journal of Finance* 46, 75-110.
- Li, Haitao, Xu Yuewu and Xiaoyan Zhang, 2004, Arbitrage and Hansen-Jagannathan Distance, unpublished working paper.
- Liang, Bing, 1999, On the Performance of Hedge Funds, *Financial Analysts Journal* 55, 72-85.
- Merton, R.C., 1981, On Market Timing and Investment Performance I: An Equilibrium Theory of Value for Market Forecasts, *Journal of Business* 54, 363-406.
- Mitchell, Mark, and Todd Pulvino, 2001, Characteristics of Risk and Return in Risk Arbitrage, *Journal of Finance* 56, 2135–2175.
- Wang, Zhenyu, and Xiaoyan Zhang, 2004, The Role of Arbitrage in the Empirical Evaluation of Asset Pricing Models, unpublished Cornell University working paper.
- Zhang, Xiaoyan, 2003, Specification Tests of International Asset Pricing Models, forthcoming in *Journal of International Money and Finance*

Table 1. Summary of empirical kernels and other factors

This table describes the construction of empirical kernels and displays correlations among kernels and other asset pricing factors. IND is the empirical kernel constructed from U.S. industry portfolios. ALT is the empirical kernel constructed from alternative assets. BOTH is the empirical kernel constructed from both industry portfolios and alternative assets. MKT, SMB, and HML are the market, size, and value factors of Fama and French. “PUT” is the return on at the money S&P 500 put options. STR is the return on at the money S&P 500 index straddles. OPUT is the return on the out-of-the-money S&P 500 index put. CYCLE is the lagged yield spread between BAA and AAA corporate bonds.

Panel A: Asset weights for empirical pricing kernels

	Non Durables	Durables	Manuf	Energy	High Tech	Telcom	Retail	Hlth	Utils	Other	Closed End	REIT	Brady	Commo dity	Global Yield
IND	19.77	9.85	3.86	-1.2	-6.6	5.31	-5.5	0.28	2.32	-27.09					
ALT											1.56	-1.82	-3.46	-0.45	5.17
BOTH	14.17	10.61	4.65	-1.55	-8.12	5.40	-3.10	2.37	-2.44	-30.68	5.39	0.47	-1.26	0.44	4.65

Panel B: Summary statistics for factors

Variable	Mean	Std	Min	Max	Correlations									
					MKT	SMB	HML	IND	ALT	BOTH	PUT	STR	OPUT	CYCLE
MKT	0.005	0.005	-0.162	0.080	1									
SMB	0.001	0.005	-0.166	0.218	0.16	1								
HML	0.004	0.005	-0.127	0.137	-0.60	-0.53	1							
IND	1.004	0.057	-0.669	2.521	-0.09	-0.14	0.25	1						
ALT	0.995	0.023	0.630	1.953	-0.29	-0.14	0.02	0.14	1					
BOTH	0.994	0.061	-0.393	2.627	-0.10	-0.16	0.03	0.84	0.37	1				
PUT	0.261	0.162	-2.609	5.768	0.73	-0.11	-0.35	-0.10	-0.16	-0.10	1			
STR	0.318	0.220	-2.882	8.948	0.50	-0.21	-0.21	-0.04	-0.03	-0.01	0.92	1		
OPUT	-0.207	0.149	-2.829	3.875	-0.85	-0.08	0.46	0.15	0.29	0.21	-0.81	-0.52	1	
CYCLE	0.000	0.000	-0.003	0.004	0.03	-0.04	0.03	0.15	0.01	0.15	0.00	-0.01	-0.01	1

Table 2. Summary statistics of hedge fund returns and characteristics

This table provides summary statistics of returns and certain fund characteristics of hedge funds in the TASS dataset. The sample period is from January 1994 to September 2003. At September 2003, the sample has 4131 funds, of which 2460 are “live” funds and 1671 are “dead” funds. “Return” is the monthly net-of-fee return based on reported net asset value. The information on size, management and incentive fees, and leverage is as of September 2003.

	All			Live			Dead		
	Mean	Median	Standard Deviation	Mean	Median	Standard Deviation	Mean	Median	Standard Deviation
Return	0.87%	0.70%	6.01%	1.00%	0.75%	5.20%	0.58%	0.57%	7.44%
Size (\$ million)	103	22	520	127	31	599	51	10	284
Management fee	1.5%	1.3%	0.8%	1.4%	1.2%	0.8%	1.6%	1.5%	1.0%
Incentive fee	16.6%	20.0%	7.3%	16.4%	20.0%	7.2%	16.9%	20.0%	7.5%
Leverage	58.6%	0.0%	204.7%	66.5%	0.0%	236.4%	41.6%	0.0%	108.2%

Table 3. Summary information on style and characteristic sorted hedge fund portfolios

This table presents summary information on portfolios of hedge funds in the TASS dataset sorted by styles and certain characteristics. Definitions of the eleven styles are given in the appendix. Market beta sorted portfolios in Panel C are constructed as follows. For each period of 24 months, beta is individually measured for each fund and, at month's end, funds are sorted into 10 portfolios according to beta. Thus, rebalancing of the portfolios occurs monthly. Volatility portfolios in Panel D are formed in a similar way as beta portfolios based on return volatilities estimated using rolling 24-month windows. For size portfolios in Panel E, each month, hedge funds are sorted into portfolios based on each fund's net asset value six months ago. For momentum portfolios in Panel F, each month hedge funds are sorted into portfolios based on their previous six months' performance. In turn, these portfolios are held for the subsequent six months. The cluster portfolios in Panel G are formed based on the clustering analysis of Ahn, Conrad, and Dittmar (2003). The cluster portfolios are formed at the end of our sample period, September 2003, based on the historical return correlation matrix. Within each panel, "mgmt" refers to the management fee.

Panel A: Style portfolio returns

	# funds	mean	stdev	t-stat	beta	volatility	size	age	mgmt	incentive	leverage
convertible arbitrage	79	0.94%	1.65%	6.15	0.13	1.69%	117	41	0.012	0.174	1.478
dedicated short bias	17	0.53%	4.68%	1.23	-1.03	6.81%	56	59	0.012	0.191	0.409
emerging markets	122	0.96%	5.08%	2.04	0.57	6.98%	68	44	0.014	0.150	0.159
equity market neutral	77	0.80%	0.53%	16.10	0.03	2.54%	98	36	0.013	0.187	0.459
event driven	178	0.92%	1.34%	7.37	0.23	2.45%	99	57	0.013	0.184	0.353
fixed income arbitrage	65	0.65%	1.15%	6.09	0.03	2.10%	118	39	0.013	0.205	3.604
fund of funds	369	0.75%	1.78%	4.54	0.17	2.71%	79	51	0.017	0.098	0.127
global macro	86	1.18%	3.77%	3.37	0.09	4.42%	145	49	0.017	0.180	1.226
long/short equity hedge	563	1.18%	2.42%	5.23	0.54	5.18%	96	42	0.011	0.183	0.341
managed futures	211	0.83%	2.94%	3.06	-0.04	6.02%	31	60	0.024	0.186	0.585
multi-strategy	45	1.13%	3.80%	3.21	0.19	3.36%	262	36	0.012	0.165	0.836

Table 3 continued.

Panel B: Correlation between styles

	Convertible arbitrage	Dedicated short bias	Emerging markets	equity market neutral	event driven	Fixed Income arbitrage	fund of funds	Global macro	long/short equity hedge	managed futures	multi- strategy
convertible arbitrage	1										
dedicated short bias	-0.37	1									
emerging markets	0.41	-0.61	1								
equity market neutral	0.34	-0.27	0.18	1							
event driven	0.59	-0.66	0.66	0.34	1						
Fixed income arbitrage	0.39	-0.07	0.24	0.11	0.38	1					
Fund of funds	0.53	-0.49	0.63	0.32	0.80	0.41	1				
global macro	0.27	-0.07	0.24	0.20	0.38	0.51	0.64	1			
long/short equity hedge	0.55	-0.63	0.56	0.38	0.79	0.20	0.82	0.34	1		
Managed futures	-0.13	0.29	-0.16	0.01	-0.19	-0.04	0.19	0.29	-0.01	1	
multi-strategy	0.27	-0.29	0.33	0.09	0.53	0.22	0.74	0.56	0.55	0.19	1.00

Panel C: Market beta sorted portfolios

Rank	mean	stdev	t-value	beta	volatility	size	age	mgmt	incentive	leverage
Low	0.95%	2.61%	3.76	-0.77	6.69%	72	52	0.019	0.185	0.568
2	0.65%	1.76%	3.81	-0.14	3.11%	102	50	0.017	0.181	0.909
3	0.78%	2.11%	3.82	-0.02	1.98%	147	46	0.015	0.168	1.072
4	0.91%	1.52%	6.14	0.05	1.92%	116	47	0.015	0.162	0.852
5	1.00%	1.54%	6.68	0.13	2.41%	97	47	0.014	0.159	0.549
6	1.10%	2.28%	4.94	0.23	2.95%	87	47	0.015	0.153	0.388
7	1.10%	2.19%	5.16	0.35	3.64%	101	50	0.014	0.158	0.330
8	1.01%	3.45%	3.01	0.52	4.55%	107	52	0.014	0.155	0.315
9	1.22%	4.16%	3.02	0.78	6.07%	69	51	0.013	0.163	0.349
High	1.50%	6.12%	2.52	1.51	10.12%	64	48	0.013	0.169	0.348
10-1	0.54%	7.61%	0.74							

Table 3 continued.**Panel D: Volatility sorted portfolios**

Rank	mean	stdev	t-value	beta	volatility	size	age	mgmt	incentive	leverage
Low	0.71%	0.76%	9.69	0.03	0.63%	137	44	0.013	0.157	1.415
2	0.75%	1.57%	4.90	0.07	1.16%	132	50	0.014	0.148	0.637
3	0.87%	1.36%	6.60	0.12	1.71%	103	51	0.014	0.152	0.536
4	1.10%	1.61%	7.00	0.17	2.31%	119	51	0.015	0.154	0.540
5	1.01%	2.04%	5.10	0.21	2.94%	99	53	0.015	0.161	0.470
6	0.99%	2.47%	4.13	0.25	3.67%	102	55	0.015	0.166	0.483
7	1.46%	2.30%	6.55	0.30	4.53%	105	53	0.016	0.171	0.447
8	0.79%	3.43%	2.38	0.33	5.67%	99	54	0.016	0.174	0.437
9	1.05%	3.73%	2.89	0.45	7.46%	70	56	0.017	0.175	0.455
High	1.88%	5.98%	3.23	0.68	12.89%	41	51	0.016	0.182	0.460
10-1	1.17%	5.69%	2.11							

Panel E: Size-sorted portfolios

rank	mean	stdev	t-value	beta	volatility	size	age	mgmt	incentive	leverage
small	1.37%	2.37%	6.07	0.24	6.59%	1	49	0.018	0.176	0.407
2	1.18%	1.97%	6.34	0.26	5.47%	3	47	0.017	0.166	0.488
3	1.27%	1.92%	6.97	0.26	4.92%	6	47	0.016	0.163	0.497
4	1.12%	2.96%	4.00	0.29	4.64%	11	46	0.016	0.163	0.535
5	1.42%	2.06%	7.24	0.28	4.46%	17	48	0.015	0.163	0.509
6	1.16%	1.95%	6.25	0.27	4.15%	26	47	0.015	0.166	0.496
7	0.96%	1.75%	5.76	0.29	3.91%	41	50	0.015	0.160	0.686
8	0.95%	1.81%	5.53	0.26	3.65%	67	54	0.014	0.163	0.747
9	1.01%	1.81%	5.89	0.24	3.43%	123	60	0.014	0.162	0.547
big	1.03%	2.11%	5.13	0.28	3.26%	635	72	0.014	0.161	0.766
10-1	-0.34%	2.14%	-1.67							

Table 3 continued.

Panel F: Momentum-sorted portfolios

rank	mean	stdev	t-value	beta	volatility	size	age	mgmt	incentive	leverage
loser	0.44%	3.30%	1.36	0.37	7.84%	66	53	0.016	0.169	0.417
2	0.70%	1.95%	3.70	0.27	4.80%	70	54	0.017	0.166	0.448
3	0.78%	1.51%	5.29	0.22	3.72%	97	53	0.016	0.162	0.526
4	0.83%	1.24%	6.84	0.19	3.01%	107	53	0.015	0.155	0.620
5	0.85%	1.24%	7.05	0.18	2.67%	107	52	0.014	0.153	0.644
6	0.87%	1.29%	6.87	0.19	2.74%	107	53	0.014	0.153	0.658
7	0.97%	1.49%	6.72	0.21	3.00%	112	53	0.015	0.159	0.675
8	1.07%	1.81%	6.04	0.27	3.70%	117	52	0.015	0.168	0.667
9	1.20%	2.37%	5.20	0.34	4.81%	111	52	0.015	0.175	0.585
winner	1.50%	3.84%	4.01	0.43	7.67%	96	49	0.016	0.181	0.488
10-1	1.06%	3.85%	2.83							

Panel G: Cluster portfolios

rank	mean	stdev	t-value	beta	volatility	size	age	mgmt	incentive	leverage
1	0.80%	1.53%	5.61	0.20	3.94%	109	41	0.015	0.177	0.656
2	0.81%	2.98%	2.93	0.44	4.26%	38	37	0.012	0.122	0.554
3	0.87%	1.11%	8.43	0.21	2.20%	93	63	0.013	0.167	0.624
4	0.95%	1.41%	7.20	0.23	4.40%	75	43	0.016	0.165	0.609
5	0.96%	2.20%	4.72	0.34	4.00%	34	53	0.011	0.162	0.951
6	0.97%	2.00%	5.25	0.14	4.77%	49	46	0.017	0.169	0.366
7	0.99%	3.85%	2.76	0.26	4.68%	148	39	0.016	0.167	0.502
8	1.04%	1.67%	6.69	0.00	4.45%	85	65	0.019	0.160	0.711
9	1.06%	3.36%	3.41	0.59	4.91%	127	57	0.012	0.151	0.244
10	1.51%	4.53%	3.60	0.34	4.70%	49	38	0.013	0.163	0.685

Table 4. Performance evaluation using the first Hansen Jagannathan (HJ) distance without arbitrage-free requirement.

This table reports summary diagnostics for the HJ-distance. The Hansen-Jagannathan distance, d , is presented in the first row. “prc err” is the maximum annual pricing error for a portfolio when its annual standard error is 20%. The error has been adjusted for small sample bias. “ $p(y<0)$ ” is the empirical p-value testing the probability that the pricing model’s SDF takes a negative value. “ $p(d=0)$ ” is the specification test, the empirical p-value for the test of $d=0$ from Monte Carlo experiments. “CAPM” denotes a one factor model while “FF” is a four factor model using the market, size, and value factors of Fama and French. “PUT” uses market factor and return on at the money index put. “OPT” uses market factor, return on an at the money straddle for index option and return on an out of money put (jump). “IND” is the empirical kernel derived from the industry portfolios, “ALT” is the kernel derived from alternative assets, and “BOTH” is the kernel derived from both sets of assets. The suffix “IV” indicates that the model includes the lagged BAA – AAA yield spread (CYCLE) as an instrumental variable.

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
11 style portfolios											
d	1.420	1.368	1.404	1.356	1.258	1.246	1.251	1.241	1.548	1.421	1.533
prc err	20%	17%	20%	18%	12%	13%	13%	14%	20%	19%	20%
$p(y<0)$	0%	8%	1%	7%	17%	17%	14%	15%	4%	0%	6%
$p(d=0)$	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
10 volatility portfolios											
d	0.935	0.850	0.890	0.808	0.931	0.849	0.890	0.819	1.007	0.945	0.954
prc err	10%	7%	9%	7%	10%	8%	9%	7%	10%	10%	8%
$p(y<0)$	0%	16%	16%	27%	2%	16%	17%	18%	4%	0%	6%
$p(d=0)$	0%	1%	0%	2%	0%	1%	0%	1%	0%	0%	0%
10 size portfolios											
d	0.549	0.547	0.440	0.440	0.368	0.357	0.324	0.314	0.660	0.611	0.681
prc err	1%	2%	-2%	-1%	-4%	-3%	-4%	-4%	2%	3%	3%
$p(y<0)$	0%	0%	27%	27%	17%	17%	19%	19%	4%	0%	6%
$p(d=0)$	28%	20%	70%	57%	91%	88%	93%	91%	17%	9%	14%
10 momentum portfolios											
d	0.654	0.526	0.630	0.477	0.617	0.507	0.616	0.500	0.666	0.691	0.646
prc err	4%	0%	4%	0%	4%	0%	4%	1%	3%	5%	2%
$p(y<0)$	0%	24%	3%	38%	5%	23%	4%	24%	4%	0%	6%
$p(d=0)$	4%	47%	4%	52%	7%	43%	5%	37%	13%	3%	21%
10 cluster portfolios											
d	0.719	0.709	0.494	0.482	0.700	0.695	0.584	0.579	0.714	0.640	0.659
prc err	6%	6%	0%	1%	6%	6%	2%	3%	4%	4%	2%
$p(y<0)$	0%	0%	14%	17%	0%	1%	25%	26%	4%	0%	6%
$p(d=0)$	1%	0%	47%	39%	1%	1%	25%	17%	7%	6%	18%

Table 5. Factor prices using the first Hansen Jagannathan (HJ) distance without arbitrage-free requirement.

Factor prices are b's defined in equation (7). "CAPM" denotes a one factor model while "FF" is a four factor model using the market, size, and value factors of Fama and French. "PUT" uses market factor and return on at the money index put. "OPT" uses market factor, return on an at the money straddle for index option, and return on an out of money put (jump). "CYCLE" is the conditional variable, calculated as the lagged BAA – AAA yield spread. CNST denotes the constant. MKT is the market factor, SMB is the size factor, HML is the B/M factor. PUT is the return on an at-the-money index put option. STR is the return on an at-the-money index straddle. OPUT is the return on an out-of-the-money index put option. The first line is the parameter estimate, and the second line is the t-value. Bold indicates significance at the 5% level.

Panel A: Factor prices for 11 style portfolios

model	CNST	MKT	SMB	HML	PUT	STR	OPUT	CYCLE
CAPM	1.01	-2.68						
	37.28	-1.04						
CAPM+IV	0.78	-1.96						990.59
	5.54	-0.78						1.81
FF	1.10	-9.31	-2.38	-13.12				
	9.29	-1.46	-0.37	-1.21				
FF+IV	0.87	-7.28	-4.21	-12.03				961.99
	4.33	-0.93	-0.52	-0.93				1.81
PUT	1.18	26.09			-1.48			
	5.47	2.59			-3.31			
PUT+IV	1.05	23.82			-1.34			489.18
	4.57	2.28			-2.99			0.76
OPT	1.20	35.93				-0.63	1.21	
	5.54	1.49				-2.93	1.25	
OPT+IV	1.08	32.08				-0.59	1.06	445.41
	4.39	1.26				-2.68	1.06	0.63

Panel B: Factor prices for 10 volatility portfolios

model	CNST	MKT	SMB	HML	PUT	STR	OPUT	CYCLE
CAPM	1.05	-8.34						
	18.15	-2.27						
CAPM+IV	0.76	-7.09						1239.11
	4.27	-1.83						1.84
FF	1.29	-22.01	-19.23	-36.33				
	4.18	-1.46	-1.01	-1.27				
FF+IV	1.00	-19.44	-17.87	-33.11				1185.16
	2.93	-1.24	-0.93	-1.16				1.82
PUT	1.07	-4.32			-0.19			
	13.29	-0.92			-0.88			
PUT+IV	0.75	-8.03			0.04			1258.12
	3.30	-0.84			0.10			1.82
OPT	1.04	-35.32				-0.38	-1.20	
	7.92	-1.69				-1.91	-1.51	
OPT+IV	0.75	-33.71				-0.23	-1.10	1163.67
	3.06	-1.12				-0.86	-0.98	1.63

Table 5 continued.

Panel C: Factor prices for 10 size portfolios

model	CNST	MKT	SMB	HML	PUT	STR	OPUT	CYCLE
CAPM	1.07	-11.60						
	14.03	-2.50						
CAPM+IV	1.03	-11.65						150.26
	10.23	-2.53						0.45
FF	0.87	-7.40	25.63	35.20				
	5.14	-1.22	1.85	1.80				
FF+IV	0.87	-7.29	25.73	35.49				-40.32
	4.81	-1.19	1.84	1.74				-0.08
PUT	1.18	8.78			-1.03			
	7.32	0.85			-2.16			
PUT+IV	1.11	9.19			-1.05			304.67
	6.58	0.92			-2.33			0.51
OPT	1.22	29.72				-0.44	1.37	
	5.86	1.15				-1.32	1.35	
OPT+IV	1.17	29.44				-0.46	1.36	266.96
	5.19	1.11				-1.41	1.30	0.34

Panel D: Factor prices for 10 momentum portfolios

model	CNST	MKT	SMB	HML	PUT	STR	OPUT	CYCLE
CAPM	1.03	-5.71						
	24.01	-1.80						
CAPM+IV	0.62	-4.56						1788.52
	1.60	-1.43						2.15
FF	1.14	-9.86	-11.20	-16.41				
	7.44	-1.28	-1.79	-1.07				
FF+IV	0.74	-12.21	-12.43	-24.58				1940.92
	1.74	-1.42	-0.84	-1.35				2.24
PUT	1.08	4.03			-0.46			
	12.27	0.73			-2.10			
PUT+IV	0.68	1.86			-0.31			1657.61
	1.79	0.25			-0.83			2.15
OPT	1.08	7.45				-0.19	0.39	
	11.16	0.61				-1.29	0.72	
OPT+IV	0.64	-7.27				-0.24	-0.27	1782.17
	1.53	-0.42				-0.98	-0.37	2.02

Table 5 continued.

Panel E: Factor prices for 10 cluster portfolios

model	CNST	MKT	SMB	HML	PUT	STR	OPUT	CYCLE
CAPM	1.00	-1.27						
	47.79	-0.50						
CAPM+IV	0.93	-1.42						327.63
	15.08	-0.55						0.98
FF	1.17	-6.05	-23.96	-23.33				
	6.12	-0.83	-2.39	-1.62				
FF+IV	1.25	-6.24	-26.15	-25.28				-320.49
	4.62	-0.87	-2.25	-1.64				-0.63
PUT	0.96	-11.82			0.48			
	18.97	-1.03			0.85			
PUT+IV	0.91	-10.65			0.42			231.02
	13.92	-0.91			0.73			0.64
OPT	0.98	17.33				0.76	1.25	
	6.04	0.83				1.62	1.53	
OPT+IV	0.94	18.29				0.73	1.27	211.01
	5.28	0.91				1.51	1.64	0.40

Table 6. Pricing errors using the first Hansen Jagannathan (HJ) distance without arbitrage-free requirement.

Pricing errors are defined in equation (6). They can be interpreted in a manner similar to a traditional Jensen's alpha. "CAPM" denotes a one factor model while "FF" is a four factor model using the market, size, and value factors of Fama and French. "PUT" uses market factor and return on at the money index put. "OPT" uses market factor, return on an at the money straddle for index option and return on an out of money put (jump). "IND" is the empirical kernel derived from the industry portfolios, "ALT" is the kernel derived from alternative assets, and "BOTH" is the kernel derived from both sets of assets. The suffix "IV" indicates that the model includes the lagged BAA – AAA yield spread (CYCLE) as an instrumental variable. Bold indicates the pricing error is significant at the 5% level.

Panel A: Pricing error for 11 style portfolios

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
convertible arbitrage	0.73%	0.44%	0.72%	0.43%	0.22%	0.12%	0.20%	0.11%	0.69%	0.67%	0.61%
dedicated short bias	0.59%	0.57%	0.48%	0.50%	0.62%	0.60%	0.55%	0.55%	0.55%	0.66%	0.61%
emerging markets	0.34%	0.78%	0.13%	0.55%	0.91%	1.07%	0.82%	0.98%	0.32%	-0.07%	-0.02%
equity market neutral	0.47%	0.44%	0.47%	0.45%	0.33%	0.33%	0.34%	0.34%	0.58%	0.49%	0.57%
event driven	0.61%	0.54%	0.55%	0.46%	0.63%	0.60%	0.57%	0.54%	0.60%	0.50%	0.49%
fixed income arbitrage	0.35%	0.50%	0.33%	0.46%	0.04%	0.14%	-0.01%	0.09%	0.26%	0.26%	0.14%
fund of funds	0.42%	0.34%	0.44%	0.33%	0.29%	0.26%	0.18%	0.17%	0.29%	0.36%	0.23%
global macro	0.90%	0.72%	0.52%	0.44%	0.25%	0.22%	-0.06%	-0.03%	0.60%	0.65%	0.11%
long/short equity hedge	0.83%	0.66%	0.96%	0.73%	0.79%	0.71%	0.75%	0.69%	0.84%	0.88%	0.86%
managed futures	0.62%	0.58%	0.61%	0.57%	0.52%	0.50%	0.45%	0.45%	0.35%	0.59%	0.39%
multi-strategy	0.73%	0.47%	0.83%	0.52%	0.15%	0.08%	-0.21%	-0.21%	0.21%	0.62%	0.14%

Panel B: Pricing error for volatility portfolios

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
Low	0.33%	0.26%	0.30%	0.22%	0.32%	0.26%	0.27%	0.22%	0.39%	0.37%	0.36%
2	0.32%	0.27%	0.08%	0.05%	0.30%	0.28%	0.46%	0.42%	0.37%	0.38%	0.31%
3	0.35%	0.30%	0.16%	0.12%	0.31%	0.31%	0.28%	0.28%	0.40%	0.43%	0.32%
4	0.42%	0.39%	0.22%	0.21%	0.42%	0.38%	0.43%	0.39%	0.67%	0.62%	0.57%
5	0.21%	-0.05%	0.04%	-0.20%	0.21%	-0.06%	0.16%	-0.08%	0.45%	0.44%	0.38%
6	0.32%	0.20%	0.38%	0.25%	0.37%	0.18%	0.59%	0.38%	0.46%	0.56%	0.40%
7	0.92%	0.51%	0.73%	0.36%	0.97%	0.50%	1.09%	0.63%	1.13%	1.05%	1.01%
8	-0.03%	-0.16%	-0.29%	-0.39%	-0.11%	-0.14%	0.23%	0.14%	0.21%	0.22%	-0.04%
9	-0.17%	0.03%	-0.35%	-0.15%	-0.10%	0.01%	-0.08%	0.02%	0.24%	0.21%	0.17%
High	0.19%	-0.16%	0.35%	0.01%	0.11%	-0.15%	0.58%	0.26%	0.96%	1.06%	0.63%

Table 6 continued.

Panel C: Pricing error for size portfolios

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
Small	0.64%	0.59%	0.09%	0.10%	0.30%	0.18%	0.30%	0.19%	0.86%	0.98%	0.93%
2	0.38%	0.40%	0.29%	0.28%	0.35%	0.39%	0.19%	0.22%	0.82%	0.68%	0.72%
3	0.56%	0.53%	0.59%	0.59%	0.49%	0.43%	0.39%	0.35%	1.03%	0.92%	1.02%
4	0.12%	0.09%	0.03%	0.03%	0.24%	0.16%	-0.03%	-0.09%	0.57%	0.61%	0.48%
5	0.45%	0.45%	0.24%	0.24%	0.06%	0.05%	-0.02%	-0.03%	0.83%	0.89%	0.80%
6	0.20%	0.17%	0.10%	0.11%	0.19%	0.14%	0.16%	0.11%	0.73%	0.67%	0.66%
7	0.09%	0.07%	0.07%	0.07%	0.04%	0.00%	0.01%	-0.03%	0.70%	0.51%	0.59%
8	0.11%	0.09%	0.10%	0.10%	0.15%	0.11%	0.10%	0.07%	0.53%	0.52%	0.46%
9	0.07%	0.05%	0.06%	0.07%	0.13%	0.08%	0.05%	0.01%	0.45%	0.49%	0.36%
Big	0.24%	0.19%	0.10%	0.11%	0.16%	0.06%	-0.05%	-0.13%	0.46%	0.53%	0.31%

Panel D: Pricing error for momentum portfolios

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
loser	-0.54%	-0.12%	-0.52%	-0.06%	-0.47%	-0.11%	-0.48%	-0.07%	-0.26%	-0.38%	-0.43%
2	-0.03%	0.20%	-0.05%	0.19%	-0.03%	0.18%	-0.04%	0.21%	0.17%	0.12%	0.09%
3	0.16%	0.28%	0.12%	0.25%	0.16%	0.27%	0.15%	0.29%	0.28%	0.26%	0.20%
4	0.26%	0.28%	0.22%	0.26%	0.21%	0.24%	0.20%	0.27%	0.32%	0.35%	0.24%
5	0.29%	0.25%	0.24%	0.20%	0.29%	0.25%	0.28%	0.28%	0.39%	0.38%	0.31%
6	0.29%	0.19%	0.24%	0.15%	0.29%	0.20%	0.27%	0.23%	0.37%	0.37%	0.30%
7	0.37%	0.24%	0.32%	0.21%	0.38%	0.26%	0.37%	0.28%	0.43%	0.47%	0.39%
8	0.39%	0.21%	0.35%	0.21%	0.42%	0.24%	0.40%	0.27%	0.46%	0.53%	0.42%
9	0.43%	0.23%	0.33%	0.18%	0.52%	0.30%	0.50%	0.34%	0.47%	0.56%	0.40%
winner	0.63%	0.34%	0.48%	0.34%	0.80%	0.47%	0.74%	0.60%	0.56%	0.76%	0.51%

Table 6 continued.

Panel E: Pricing error for cluster portfolios

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
1	0.69%	0.63%	0.46%	0.50%	0.61%	0.58%	0.42%	0.39%	0.66%	0.61%	0.59%
2	0.53%	0.52%	0.45%	0.45%	0.59%	0.57%	0.27%	0.26%	0.40%	0.38%	0.32%
3	0.62%	0.49%	0.33%	0.44%	0.65%	0.55%	0.72%	0.63%	0.58%	0.47%	0.54%
4	0.74%	0.67%	0.75%	0.83%	0.92%	0.85%	0.24%	0.17%	0.43%	0.52%	0.03%
5	1.22%	1.00%	0.30%	0.42%	0.84%	0.73%	0.58%	0.48%	0.79%	1.14%	0.91%
6	0.74%	0.63%	0.34%	0.40%	0.67%	0.60%	0.36%	0.30%	0.36%	0.49%	0.27%
7	0.64%	0.64%	0.23%	0.20%	0.63%	0.62%	0.45%	0.45%	0.62%	0.48%	0.47%
8	0.77%	0.74%	0.51%	0.51%	0.76%	0.74%	0.48%	0.46%	0.54%	0.68%	0.53%
9	0.55%	0.58%	0.70%	0.68%	0.49%	0.52%	0.56%	0.58%	0.49%	0.42%	0.46%
10	0.69%	0.66%	0.31%	0.30%	0.59%	0.58%	0.24%	0.23%	0.49%	0.59%	0.49%

Table 7. Performance evaluation based on the second Hansen Jagannathan (HJ) distance with arbitrage-free requirement.

This table reports summary diagnostics for the HJ-distance with the no-arbitrage constraint imposed. The Hansen-Jagannathan distance with the no-arbitrage constraint, d^+ , is presented in the first row. “prc err” is the maximum annual pricing error for a portfolio when its annual standard error is 20%. The error has been adjusted for small sample bias. “p(y<0)” is the empirical p-value testing the probability that the pricing model’s SDF takes a negative value. “p($d^+=0$)” is the specification test, the empirical p-value for the test of $d^+=0$ from Monte Carlo experiments. “CAPM” denotes a one factor model while “FF” is a four factor model using the market, size, and value factors of Fama and French. “PUT” uses market factor and return on at the money index put. “OPT” uses market factor, return on an at the money straddle for index option and return on an out of money put (jump). “IND” is the empirical kernel derived from the industry portfolios, “ALT” is the kernel derived from alternative assets, and “BOTH” is the kernel derived from both sets of assets. The suffix “IV” indicates that the model includes the lagged BAA – AAA yield spread (CYCLE) as an instrumental variable.

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
11 style portfolios											
d^+	2.104	2.086	2.095	2.075	2.066	2.051	2.066	2.051	2.420	2.256	2.387
prc err	34%	33%	34%	34%	33%	32%	33%	33%	37%	36%	36%
p(y<0)	0%	0%	0%	1%	3%	4%	3%	4%	4%	0%	6%
p($d^+=0$)	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
10 volatility portfolios											
d^+	1.004	0.961	0.969	0.936	1.002	0.958	1.002	0.958	1.088	0.999	1.046
prc err	11%	10%	11%	11%	12%	11%	12%	11%	11%	11%	10%
p(y<0)	0%	2%	3%	3%	0%	1%	0%	2%	4%	0%	6%
p($d^+=0$)	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
10 size portfolios											
d^+	0.611	0.609	0.597	0.592	0.586	0.584	0.584	0.582	0.723	0.672	0.736
prc err	2%	3%	3%	3%	2%	3%	3%	3%	3%	4%	3%
p(y<0)	0%	0%	3%	4%	3%	3%	3%	3%	4%	0%	6%
p($d^+=0$)	19%	15%	15%	13%	18%	17%	17%	15%	12%	4%	12%
10 momentum portfolios											
d^+	0.667	0.624	0.637	0.596	0.644	0.605	0.639	0.591	0.716	0.703	0.703
prc err	4%	4%	4%	4%	4%	4%	5%	4%	3%	5%	3%
p(y<0)	0%	2%	0%	1%	2%	4%	2%	5%	4%	0%	6%
p($d^+=0$)	4%	7%	6%	8%	3%	8%	4%	9%	10%	3%	16%
10 cluster portfolios											
d^+	0.721	0.717	0.571	0.567	0.708	0.706	0.648	0.648	0.729	0.645	0.681
prc err	6%	6%	2%	3%	6%	6%	5%	5%	3%	4%	2%
p(y<0)	0%	0%	5%	8%	0%	0%	4%	4%	4%	0%	6%
p($d^+=0$)	1%	1%	20%	18%	2%	1%	4%	3%	10%	6%	20%

Table 8. Factor prices using the second Hansen Jagannathan (HJ) distance with arbitrage-free requirement.

Factor prices are b's defined in equation (7). "CAPM" denotes a one factor model while "FF" is a four factor model using the market, size, and value factors of Fama and French. "PUT" uses market factor and return on at the money index put. "OPT" uses market factor, return on an at the money straddle for index option, and return on an out of money put (jump). "CYCLE" is the conditional variable, calculated as the lagged BAA – AAA yield spread. CNST denotes the constant. MKT is the market factor, SMB is the size factor, HML is the B/M factor. PUT is the return on an at-the-money index put option. STR is the return on an at-the-money index straddle. OPUT is the return on an out-of-the-money index put option. The first line is the parameter estimate, and the second line is the t-value. Bold indicates significance at the 5% level.

Panel A: Factor prices for 11 style portfolios

Model	CNST	MKT	SMB	HML	PUT	STR	OPUT	CYCLE
CAPM	0.99	0.96						
	89.95	0.40						
CAPM+IV	0.92	0.57						332.72
	17.60	0.23						1.58
FF	1.01	-1.44	2.15	-3.98				
	18.52	-0.37	0.42	-0.65				
FF+IV	0.93	-1.70	3.34	-3.26				355.74
	13.72	-0.47	0.62	-0.56				1.64
PUT	1.03	9.68			-0.38			
	18.19	3.45			-4.71			
PUT+IV	0.96	9.04			-0.37			307.87
	13.81	3.20			-4.62			1.44
OPT	1.03	9.85				-0.19	0.20	
	18.34	2.15				-3.12	1.21	
OPT+IV	0.96	9.01				-0.18	0.18	307.80
	13.74	2.02				-3.14	1.11	1.44

Panel B: Factor prices for 10 volatility portfolios

Model	CNST	MKT	SMB	HML	PUT	STR	OPUT	CYCLE
CAPM	1.03	-6.10						
	23.80	-2.10						
CAPM+IV	0.93	-6.58						459.51
	3.86	-2.04						0.47
FF	1.14	-11.59	-10.51	-16.27				
	13.17	-2.46	-2.16	-2.19				
FF+IV	1.05	-11.60	-9.35	-15.05				390.99
	6.35	-2.52	-1.57	-1.35				0.65
PUT	1.02	-8.45			0.11			
	18.79	-1.12			0.31			
PUT+IV	0.92	-9.09			0.11			460.56
	7.64	-1.12			0.29			0.82
OPT	1.02	-9.81				0.04	-0.11	
	19.70	-0.89				0.21	-0.32	
OPT+IV	0.92	-9.66				0.05	-0.08	458.34
	7.56	-0.70				0.27	-0.18	0.76

Table 8 continued.

Panel C: Factor prices for 10 size portfolios

Model	CNST	MKT	SMB	HML	PUT	STR	OPUT	CYCLE
CAPM	1.06	-10.70						
	14.85	-2.31						
CAPM+IV	1.03	-10.91						150.26
	7.50	-2.48						0.31
FF	1.01	-7.82	3.70	8.88				
	9.75	-1.49	0.61	1.11				
FF+IV	0.97	-7.99	4.85	10.13				131.99
	4.01	-1.54	0.57	1.01				0.18
PUT	1.08	-5.93			-0.22			
	13.22	-1.27			-2.97			
PUT+IV	1.06	-6.14			-0.21			89.58
	7.20	-1.33			-2.99			0.18
OPT	1.09	-3.57				-0.08	0.22	
	13.45	-0.35				-0.82	0.59	
OPT+IV	1.07	-3.92				-0.08	0.21	85.37
	7.12	-0.40				-0.84	0.60	0.17

Panel D: Factor prices for 10 momentum portfolios

Model	CNST	MKT	SMB	HML	PUT	STR	OPUT	CYCLE
CAPM	1.03	-5.57						
	24.32	-1.69						
CAPM+IV	0.93	-5.84						461.72
	7.82	-1.81						1.50
FF	1.11	-8.85	-8.79	-12.20				
	12.93	-1.70	-1.34	-1.67				
FF+IV	1.00	-9.48	-7.92	-12.77				503.01
	6.69	-2.11	-1.74	-1.73				0.85
PUT	1.05	-0.67			-0.22			
	18.91	-0.17			-2.52			
PUT+IV	0.95	-1.31			-0.20			448.87
	7.59	-0.36			-2.28			1.39
OPT	1.06	3.79				-0.07	0.32	
	17.16	0.42				-0.55	0.79	
OPT+IV	0.94	5.30				-0.04	0.40	556.53
	7.45	0.94				-0.69	1.85	2.42

Table 8 continued.

Panel E: Factor prices for 10 cluster portfolios

Model	CNST	MKT	SMB	HML	PUT	STR	OPUT	CYCLE
CAPM	1.00	-1.31						
	43.67	-0.46						
CAPM+IV	0.97	-1.42						150.51
	11.51	-0.50						0.34
FF	1.11	-4.41	-16.84	-15.08				
	9.14	-0.77	-3.04	-1.81				
FF+IV	1.14	-4.41	-17.44	-15.54				-119.12
	5.77	-0.77	-2.96	-2.19				-0.13
PUT	0.97	-8.55			0.33			
	21.24	-1.07			0.84			
PUT+IV	0.95	-7.90			0.29			119.62
	11.24	-0.82			0.62			0.22
OPT	0.99	6.95				0.36	0.58	
	11.82	0.58				2.06	1.34	
OPT+IV	0.99	6.96				0.36	0.58	-2.63
	4.95	0.58				1.53	1.25	0.00

Table 9. Pricing errors using the second Hansen Jagannathan (HJ) distance with arbitrage-free constraint.

Pricing errors are defined in equation (6). They can be interpreted in a manner similar to a traditional Jensen's alpha. "CAPM" denotes a one factor model while "FF" is a four factor model using the market, size, and value factors of Fama and French. "PUT" uses market factor and return on at the money index put. "OPT" uses market factor, return on an at the money straddle for index option and return on an out of money put (jump). "IND" is the empirical kernel derived from the industry portfolios, "ALT" is the kernel derived from alternative assets, and "BOTH" is the kernel derived from both sets of assets. The suffix "IV" indicates that the model includes the lagged BAA – AAA yield spread (CYCLE) as an instrumental variable. Bold indicates the pricing error is significant at the 5% level.

Panel A: Pricing error for 11 style portfolios

	CAPM	CAPM+ IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
convertible arbitrage	0.81%	0.70%	0.84%	0.73%	0.71%	0.61%	0.71%	0.61%	0.69%	0.67%	0.61%
dedicated short bias	-0.13%	-0.01%	-0.37%	-0.30%	-0.38%	-0.27%	-0.38%	-0.27%	0.55%	0.66%	0.61%
emerging markets	0.91%	0.96%	0.99%	1.10%	1.26%	1.30%	1.26%	1.30%	0.32%	-0.07%	-0.02%
equity market neutral	0.52%	0.50%	0.51%	0.49%	0.50%	0.48%	0.50%	0.48%	0.58%	0.49%	0.57%
event driven	0.78%	0.73%	0.83%	0.80%	0.84%	0.79%	0.84%	0.79%	0.60%	0.50%	0.49%
fixed income arbitrage	0.36%	0.41%	0.37%	0.43%	0.28%	0.33%	0.28%	0.33%	0.26%	0.26%	0.14%
fund of funds	0.60%	0.54%	0.67%	0.63%	0.63%	0.57%	0.63%	0.57%	0.29%	0.36%	0.23%
global macro	1.00%	0.92%	0.82%	0.74%	0.87%	0.80%	0.86%	0.80%	0.60%	0.65%	0.11%
long/short equity hedge	1.11%	1.00%	1.27%	1.19%	1.19%	1.10%	1.19%	1.10%	0.84%	0.88%	0.86%
Managed futures	0.48%	0.49%	0.45%	0.45%	0.41%	0.42%	0.41%	0.42%	0.35%	0.59%	0.39%
multi-strategy	0.99%	0.86%	1.15%	1.04%	0.94%	0.82%	0.93%	0.82%	0.21%	0.62%	0.14%

Panel B: Pricing error for volatility portfolios

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
low	0.35%	0.32%	0.33%	0.30%	0.36%	0.32%	0.36%	0.32%	0.39%	0.37%	0.36%
2	0.36%	0.33%	0.24%	0.23%	0.37%	0.34%	0.38%	0.35%	0.37%	0.38%	0.31%
3	0.42%	0.37%	0.30%	0.28%	0.44%	0.39%	0.44%	0.39%	0.40%	0.43%	0.32%
4	0.52%	0.47%	0.42%	0.39%	0.51%	0.46%	0.52%	0.46%	0.67%	0.62%	0.57%
5	0.34%	0.19%	0.23%	0.12%	0.33%	0.18%	0.33%	0.18%	0.45%	0.44%	0.38%
6	0.43%	0.34%	0.44%	0.37%	0.40%	0.30%	0.41%	0.31%	0.46%	0.56%	0.40%
7	0.99%	0.81%	0.88%	0.74%	0.96%	0.77%	0.96%	0.78%	1.13%	1.05%	1.01%
8	0.12%	0.01%	0.01%	-0.07%	0.16%	0.05%	0.18%	0.06%	0.21%	0.22%	-0.04%
9	0.07%	0.04%	-0.09%	-0.07%	0.01%	-0.01%	0.02%	-0.01%	0.24%	0.21%	0.17%
high	0.59%	0.29%	0.62%	0.39%	0.61%	0.32%	0.65%	0.34%	0.96%	1.06%	0.63%

Table 9 continued.

Panel C: Pricing error for size portfolios

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
small	0.68%	0.61%	0.53%	0.46%	0.62%	0.58%	0.63%	0.59%	0.86%	0.98%	0.93%
2	0.42%	0.43%	0.43%	0.44%	0.43%	0.44%	0.40%	0.41%	0.82%	0.68%	0.72%
3	0.60%	0.57%	0.58%	0.56%	0.61%	0.59%	0.59%	0.57%	1.03%	0.92%	1.02%
4	0.18%	0.13%	0.16%	0.12%	0.23%	0.20%	0.18%	0.16%	0.57%	0.61%	0.48%
5	0.50%	0.49%	0.45%	0.43%	0.45%	0.44%	0.44%	0.43%	0.83%	0.89%	0.80%
6	0.25%	0.22%	0.23%	0.20%	0.28%	0.25%	0.27%	0.25%	0.73%	0.67%	0.66%
7	0.14%	0.11%	0.15%	0.13%	0.15%	0.14%	0.15%	0.13%	0.70%	0.51%	0.59%
8	0.16%	0.13%	0.13%	0.11%	0.19%	0.17%	0.18%	0.16%	0.53%	0.52%	0.46%
9	0.12%	0.09%	0.11%	0.08%	0.16%	0.14%	0.14%	0.12%	0.45%	0.49%	0.36%
big	0.28%	0.23%	0.26%	0.21%	0.28%	0.25%	0.25%	0.22%	0.46%	0.53%	0.31%

Panel D: Pricing error for momentum portfolios

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
loser	-0.53%	-0.48%	-0.55%	-0.47%	-0.46%	-0.42%	-0.48%	-0.40%	-0.26%	-0.38%	-0.43%
2	-0.02%	0.00%	-0.06%	-0.02%	0.00%	0.02%	-0.02%	0.02%	0.17%	0.12%	0.09%
3	0.16%	0.17%	0.11%	0.13%	0.18%	0.18%	0.17%	0.18%	0.28%	0.26%	0.20%
4	0.26%	0.25%	0.22%	0.22%	0.25%	0.24%	0.23%	0.22%	0.32%	0.35%	0.24%
5	0.30%	0.26%	0.24%	0.22%	0.31%	0.27%	0.29%	0.25%	0.39%	0.38%	0.31%
6	0.30%	0.25%	0.24%	0.21%	0.31%	0.26%	0.28%	0.23%	0.37%	0.37%	0.30%
7	0.37%	0.31%	0.31%	0.27%	0.39%	0.34%	0.37%	0.30%	0.43%	0.47%	0.39%
8	0.40%	0.32%	0.34%	0.29%	0.43%	0.35%	0.40%	0.30%	0.46%	0.53%	0.42%
9	0.44%	0.35%	0.32%	0.27%	0.50%	0.41%	0.47%	0.36%	0.47%	0.56%	0.40%
winner	0.64%	0.51%	0.47%	0.42%	0.75%	0.62%	0.66%	0.49%	0.56%	0.76%	0.51%

Table 9 continued.

Panel E: Pricing error for cluster portfolios

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH
1	0.69%	0.66%	0.52%	0.53%	0.64%	0.62%	0.55%	0.55%	0.66%	0.61%	0.59%
2	0.53%	0.52%	0.46%	0.46%	0.57%	0.56%	0.40%	0.40%	0.40%	0.38%	0.32%
3	0.62%	0.56%	0.40%	0.44%	0.64%	0.59%	0.66%	0.66%	0.58%	0.47%	0.54%
4	0.74%	0.70%	0.77%	0.80%	0.86%	0.82%	0.50%	0.50%	0.43%	0.52%	0.03%
5	1.22%	1.11%	0.43%	0.48%	0.96%	0.90%	0.89%	0.89%	0.79%	1.14%	0.91%
6	0.73%	0.68%	0.38%	0.40%	0.68%	0.64%	0.52%	0.52%	0.36%	0.49%	0.27%
7	0.64%	0.64%	0.34%	0.33%	0.63%	0.63%	0.54%	0.54%	0.62%	0.48%	0.47%
8	0.77%	0.75%	0.58%	0.58%	0.76%	0.75%	0.63%	0.63%	0.54%	0.68%	0.53%
9	0.55%	0.56%	0.57%	0.56%	0.50%	0.51%	0.52%	0.52%	0.49%	0.42%	0.46%
10	0.68%	0.67%	0.38%	0.38%	0.62%	0.61%	0.45%	0.45%	0.49%	0.59%	0.49%

Table 10. Constructing an empirical kernel from the hedge fund style portfolios.

We construct an empirical kernel, “STYLE”, from the style portfolios to further study the mispricings that arise from the use of our other model SDFs. Panel A describes the construction of the kernel while Panel B displays correlations between STYLE and other kernels previously described. “CAPM” denotes a one factor model while “FF” is a four factor model using the market, size, and value factors of Fama and French. “PUT” uses market factor and return on at the money index put. “OPT” uses market factor, return on an at the money straddle for index option, and return on an out of money put (jump). “IND” is the empirical kernel derived from the industry portfolios, “ALT” is the kernel derived from alternative assets, and “BOTH” is the kernel derived from both sets of assets. The suffix “IV” indicates that the model includes the lagged BAA – AAA yield spread (CYCLE) as an instrumental variable.

Panel A: Asset weights for “STYLE” empirical pricing kernel

convertible arbitrage	dedicated short bias	emerging markets	equity market neutral	event driven	fixed income arbitrage	fund of funds	global macro	long/short equity hedge	managed futures	multi- strategy
0.056	0.071	0.007	0.683	0.322	0.115	-0.560	0.001	0.176	0.056	0.084

Panel B: Summary statistics for SDFs

	CAPM	CAPM+IV	FF	FF+IV	PUT	PUT+IV	OPT	OPT+IV	IND	ALT	BOTH	STYLE
CAPM	1.00											
CAPM+IV	0.08	1.00										
FF	0.36	-0.01	1.00									
FF+IV	0.08	0.94	0.31	1.00								
PUT	0.23	0.04	0.14	0.03	1.00							
PUT+IV	0.21	0.35	0.12	0.32	0.95	1.00						
OPT	0.22	0.04	0.15	0.04	0.98	0.93	1.00					
OPT+IV	0.20	0.32	0.13	0.30	0.94	0.98	0.96	1.00				
IND	0.11	0.18	-0.16	0.11	0.00	0.05	0.03	0.07	1.00			
ALT	0.24	0.09	0.23	0.15	-0.10	-0.08	-0.06	-0.05	0.47	1.00		
BOTH	0.12	0.18	0.11	0.21	-0.01	0.04	0.05	0.09	0.88	0.57	1.00	
STYLE	0.07	0.11	0.09	0.13	0.19	0.22	0.20	0.22	-0.19	0.01	-0.11	1.00

Table 11. Fund and fund manager characteristics related to fund performance

This table presents associations between individual hedge fund performance and fund characteristics including manager characteristics. Hedge fund performance is measured with raw return, excess return from the STYLE empirical kernel (STYLE error) described in the previous table and excess return from the OPT kernel (OPT error) described in Table 1. “SAT score” is the mean SAT score for students admitted to the university where the lead manager earned his or her undergraduate degree.

Panel A: Fund manager characteristics

	nobs	mean	max	75%	50%	25%	minimum
SAT score	1361	1305	1520	1410	1315	1206	865
Lead manager’s birth year	949	1958	1977	1964	1960	1953	1914
Lead manager’s tenure starts (year)	1899	1994	2003	1998	1996	1992	1961
Lead manager’s experience starts (year)	2095	1982	2002	1988	1984	1978	1950
Number of managers	1715	3	15	3	2	2	1
Dummy for lead manager holds MBA	3160	27%					
Dummy for lead manager holds CFA	3160	11%					
Dummy for lead manager holds Ivy League first degree	3160	15%					

Panel B: Correlations among fund manager characteristics, other fund characteristics, and fund performance.

	raw return	OPT error	STYLE error	SAT	Ivy dum	MBA dum	CFA dum	manager age	manager exp	manager tenure	# managers	incent fee	lev	mgmt fee
raw return	1.00													
OPT error	0.90	1.00												
STYLE error	0.72	0.66	1.00											
SAT	0.01	0.01	0.01	1.00										
Ivy dummy	0.01	0.01	0.01	0.70	1.00									
MBA dummy	0.01	0.01	0.00	0.05	0.15	1.00								
CFA dummy	0.00	0.01	0.01	-0.10	0.00	0.09	1.00							
Manager age	-0.04	-0.03	-0.02	-0.01	-0.03	0.14	-0.08	1.00						
Manager exp	0.00	0.01	0.01	0.05	0.16	0.29	0.07	0.69	1.00					
Manager tenure	-0.01	0.00	0.00	0.09	0.14	0.19	0.02	0.46	0.51	1.00				
# managers	0.01	0.01	0.01	-0.01	0.02	0.02	-0.01	-0.05	-0.05	-0.02	1.00			
Incentive fee	0.02	0.02	0.02	0.04	0.11	0.11	0.05	-0.10	0.08	0.06	-0.08	1.00		
Leverage	0.00	0.00	0.00	-0.04	-0.01	0.06	-0.02	0.06	0.09	0.06	0.03	0.13	1.00	
Mgmt fee	0.00	-0.02	-0.01	0.02	-0.04	-0.11	-0.09	-0.06	-0.14	-0.10	-0.01	0.05	-0.01	1.00

Table 12. Regressions relating fund performance to manager characteristics and other fund characteristics

This table presents associations between individual hedge fund performance and fund characteristics including manager characteristics. Hedge fund performance is measured with raw return, excess return from the STYLE empirical kernel (“STYLE error”) described in the previous table and excess return from the OPT kernel (“OPT error”) described in Table 1. Manager characteristics are defined and summarized in Table 11 while fund characteristics are summarized in Table 2. Each regression observation represents one fund-month and pools all observations for all funds. Bold indicates p-value reported beneath each coefficient is significant at the 5% level.

	Nobs	Adjusted R ²	Intercept	SAT	IVY dummy	Manager age	Manager tenure	Manager experience	MBA dummy	CFA dummy	Incentive fee	Leverage	Mgmt fee
raw return	26972	0.14%	0.01207	0.00000		-0.00017	-0.00016	-0.00003	0.00051	-0.00131	0.00599	-0.00005	0.07794
			1%	10%		0%	3%	57%	53%	25%	44%	78%	26%
raw return	26973	0.11%	0.00577	0.00001			-0.00021	-0.00014	0.00062	-0.00101	0.00690	-0.00007	0.07957
			16%	8%			0%	0%	45%	37%	37%	72%	25%
OPT error	26972	0.09%	0.00899	0.00001		-0.00011	-0.00019	-0.00006	0.00082	-0.00097	0.00492	-0.00013	0.01258
			9%	12%		9%	2%	35%	36%	44%	57%	55%	87%
OPT error	26973	0.08%	0.00485	0.00001			-0.00022	-0.00014	0.00090	-0.00077	0.00552	-0.00014	0.01365
			30%	10%			0%	1%	32%	54%	52%	52%	86%
STYLE error	26972	0.01%	0.00944	0.00000		-0.00014	-0.00024	0.00002	0.00004	-0.00006	0.00741	-0.00027	-0.08458
			29%	38%		21%	7%	87%	98%	98%	62%	46%	52%
STYLE error	26973	0.01%	0.00423	0.00001			-0.00029	-0.00007	0.00013	0.00019	0.00816	-0.00029	-0.08323
			59%	35%			3%	42%	93%	93%	58%	44%	52%
raw return	26972	0.15%	0.01782		0.00183	-0.00018	-0.00016	-0.00003	0.00063	-0.00127	0.00555	-0.00006	0.07650
			0%		3%	0%	2%	67%	44%	26%	47%	77%	26%
raw return	26973	0.12%	0.01184		0.00182		-0.00021	-0.00014	0.00075	-0.00098	0.00655	-0.00008	0.07796
			0%		3%		0%	0%	36%	39%	40%	70%	26%
OPT error	26972	0.09%	0.01511		0.00205	-0.00012	-0.00019	-0.00005	0.00095	-0.00092	0.00440	-0.00013	0.01110
			0%		3%	8%	2%	42%	29%	47%	61%	54%	88%
OPT error	26973	0.09%	0.01111		0.00204		-0.00023	-0.00013	0.00103	-0.00072	0.00507	-0.00015	0.01208
			0%		3%		0%	1%	25%	57%	56%	50%	87%
STYLE error	26972	0.01%	0.01544		0.00185	-0.00015	-0.00025	0.00003	0.00016	-0.00003	0.00698	-0.00028	-0.08610
			0%		24%	20%	7%	81%	92%	99%	64%	45%	51%
STYLE error	26973	0.01%	0.01046		0.00184		-0.00029	-0.00007	0.00026	0.00022	0.00781	-0.00029	-0.08488
			0%		25%		3%	46%	87%	92%	60%	43%	52%